External Dependencies for Java

References

- jqassistant
- Neo4j Python Driver

External Package Usage

External Package

An external type has no `byteCodeVersion` since it only occurs as a dependency but wasn't analyzed itself (missing bytecode). Core Java types like `java.lang.Integer` and primitives like `int` are considered "build-in" and therefore aren't interpreted as "external" even though their byte code is also missing. A package is categorized as "external" if the types it contains are classified as external.

External annotation dependency

The aforementioned classification encompasses external annotation dependencies as well. These dependencies introduce significantly less coupling and are not indispensable for compiling code. Without the external annotation the code would most probably behave differently. Hence, they are included in the first more overall and general tables and then left out in the later more specific ones.

Table 1 - Top 20 most used external packages overall

This table shows the external packages that are used by the most different internal types overall. Additionally, it shows which types of the external package are actually used. External annotations are also listed.

Only the top 20 entries are shown. The whole table can be found in the following CSV report: `External_package_usage_overall`

Columns:

- *externalPackageName* identifies the external package as described above
- *numberOfExternalCallerPackages* refers to the distinct packages that make use of the external package
- *numberOfExternalCallerTypes* refers to the distinct types that make use of the external package
- *numberOfExternalTypeCalls* includes every dependency to the types in the external package
- *numberOfExternalTypeCallsWeighted* includes every invocation or reference (sum of weights) to the types in the external package
- *allPackages* contains the total count of all analyzed packages in general
- *allTypes* contains the total count of all analyzed types in general
- *externalTypeNames* contains a list of actually utilized types of the external package

	externalPackageName	numberOfExternalCallerPackages	numberOfExternalCallerTypes	numberOfExternalTypeCalls	number
0	jakarta.annotation	102	684	870	
1	org.slf4j	38	74	133	
2	io.axoniq.axonserver.connector	5	16	23	
3	io.axoniq.axonserver.grpc	5	18	40	
4	jakarta.persistence	5	18	42	
5	reactor.core.publisher	5	8	14	
6	com.fasterxml.jackson.annotation	3	6	18	
7	com.fasterxml.jackson.databind	3	9	14	
8	io.grpc	3	7	23	
9	org.hamcrest	3	22	44	
10	org.reactivestreams	3	6	8	
11	org.springframework.boot.actuate.health	3	4	7	
12	com.fasterxml.jackson.databind.node	2	3	5	
13	io.axoniq.axonserver.grpc.control	2	6	13	
14	io.grpc.stub	2	3	4	
15	io.micrometer.core.instrument	2	12	38	
16	org.apache.avro	2	5	17	
17	org.apache.avro.message	2	9	12	
18	org.axonframework.extension.spring.config	2	5	13	
19	org.springframework.boot.autoconfigure	2	20	30	

Table 1 Chart 1a - Most called external packages in % by types (more than 0.7% overall)

External packages that are used less than 0.7% are grouped into the name "others" to get a cleaner chart with the most significant external packages and how ofter they are called in percent.

```
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```

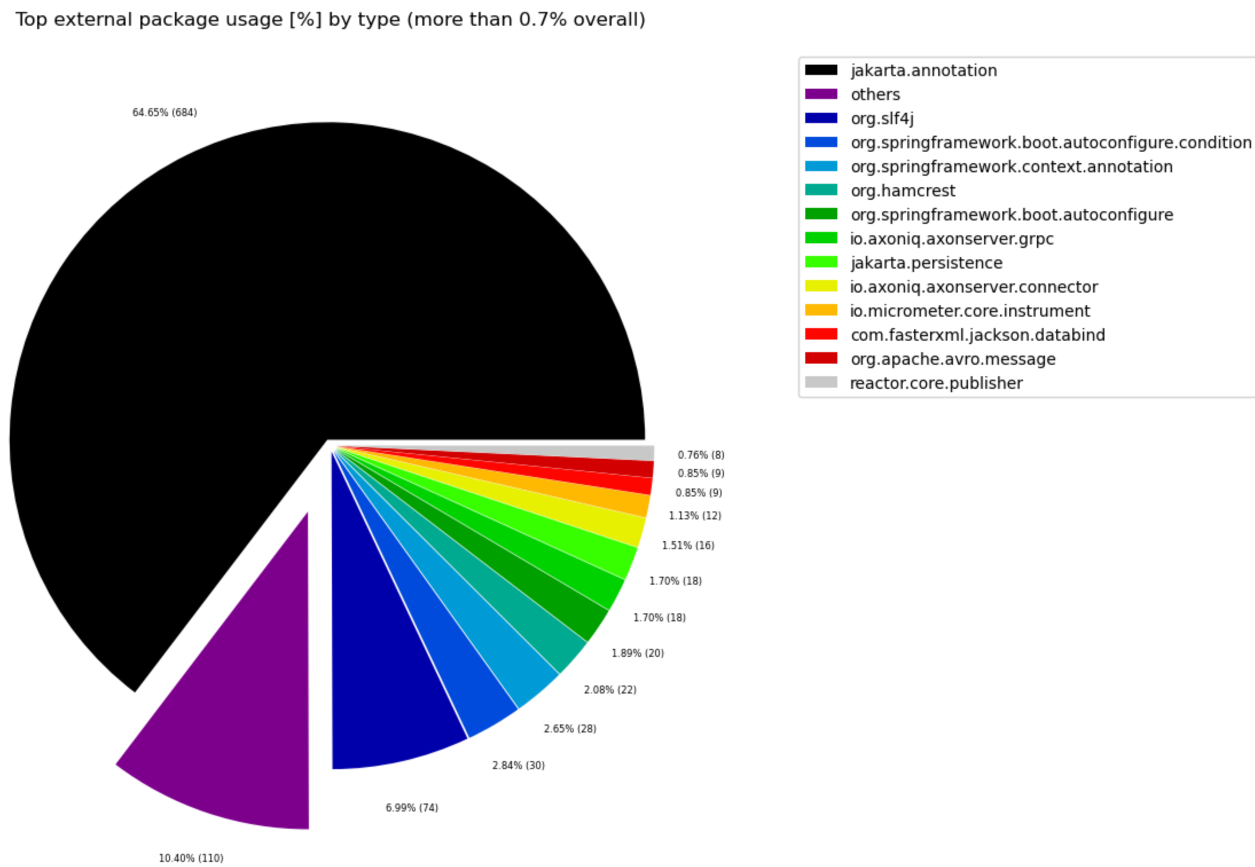

Table 1 Chart 1b - Most called external packages in % by types (less than 0.7% overall "others" drill-down)

Shows the lowest (less than 0.7% overall) most called external package. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.3% from that will be grouped into "others" to get a cleaner plot.

```
<Figure size 640x480 with 0 Axes>
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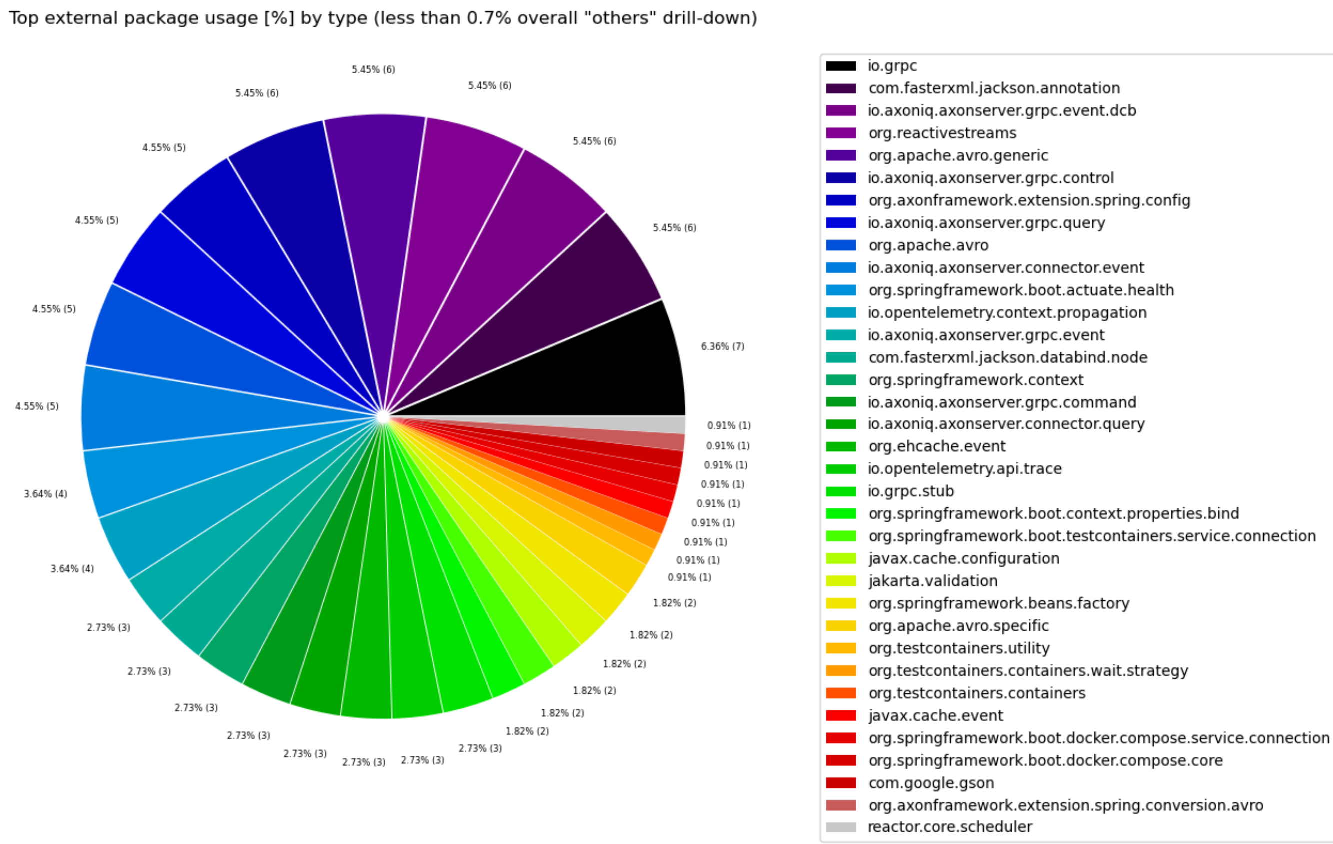

Table 1 Chart 2a - Most called external packages in % by packages (more than 0.7% overall)

External packages that are used less than 0.7% are grouped into the name "others" to get a cleaner chart with the most significant external packages and how ofter they are called in percent.

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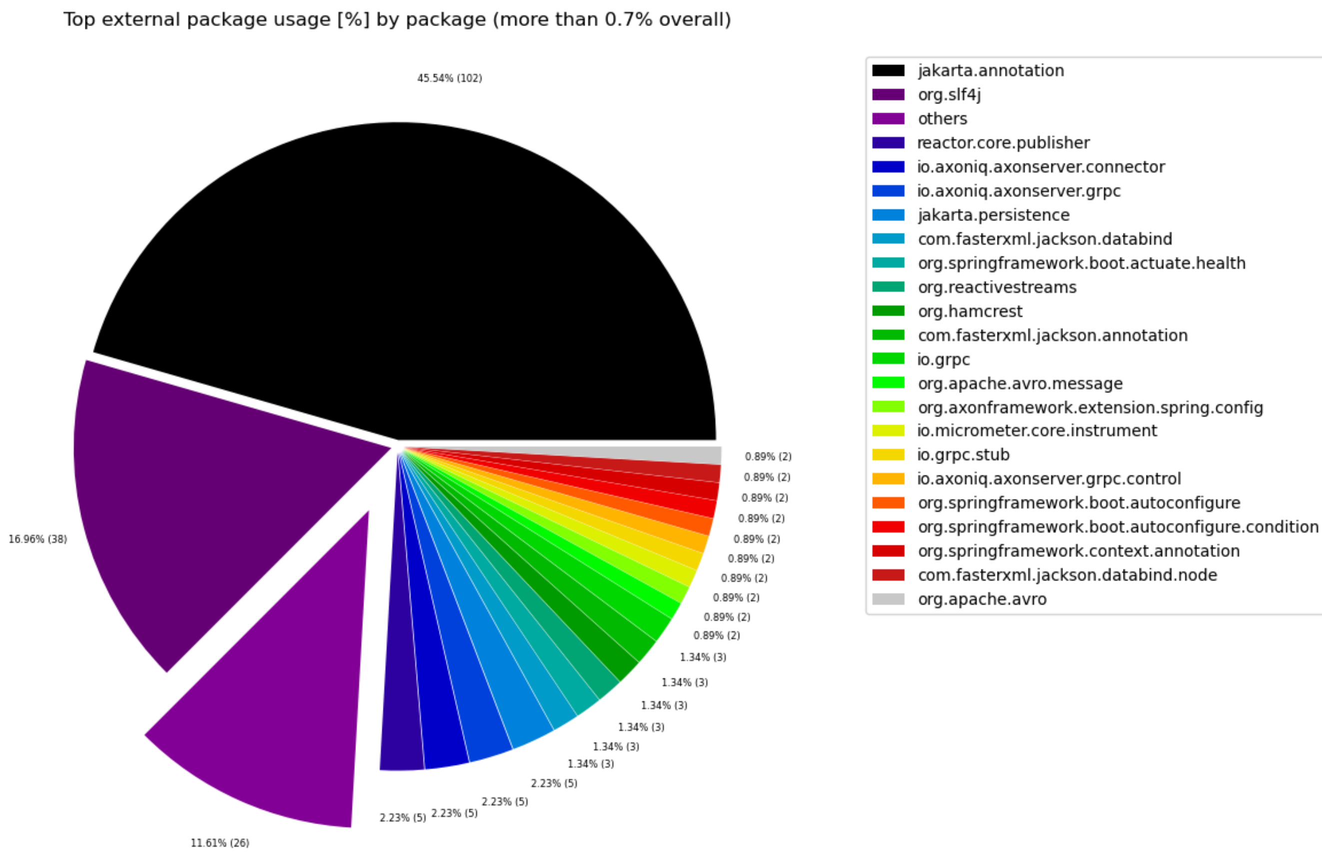

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```

Top external package usage [%] by package (less than 0.7% overall "others" drill-down)

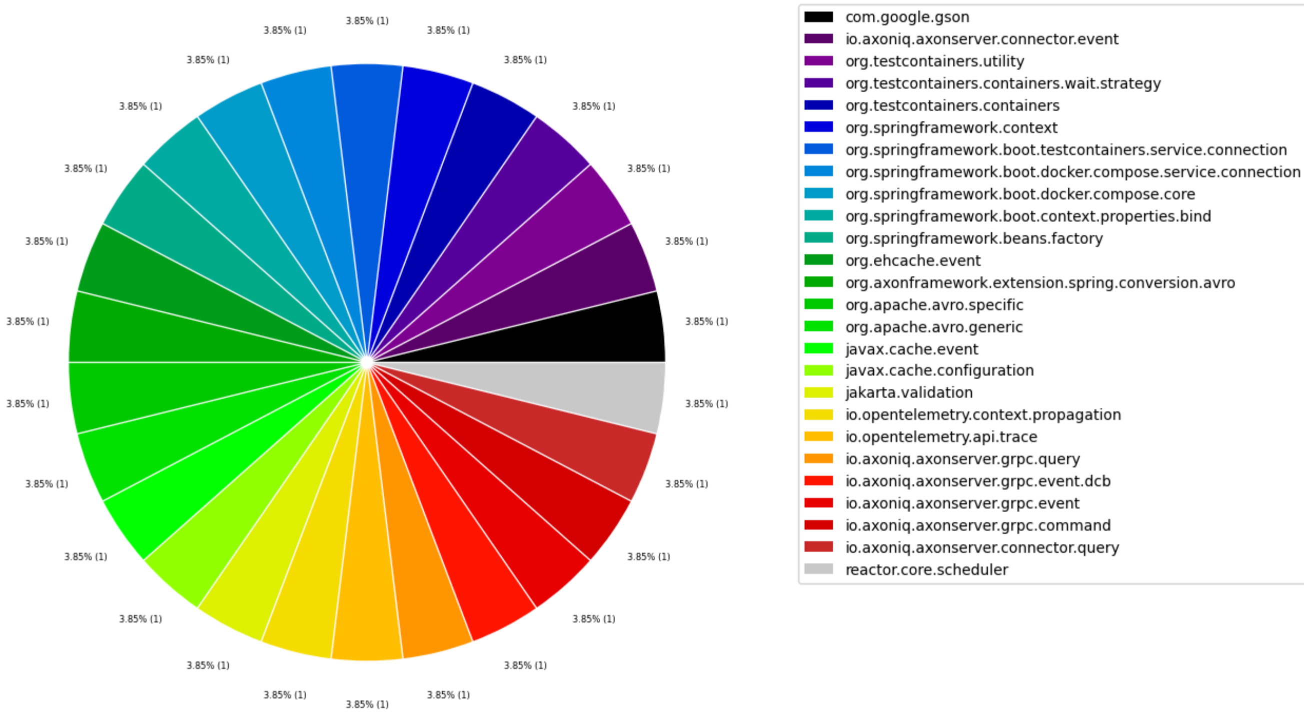

Table 2 - Top 20 most used external packages grouped by their first 2 layers

This table shows external packages grouped by their first 2 layers that are used by the most different internal types overall including external annotations. For example, "javax.xml.stream" and "javax.xml.parsers" are grouped together to "javax.xml".

Additionally, it shows which types of the external packages are actually used.

Only the top 20 entries are shown. The whole table can be found in the following CSV report: `External_second_level_package_usage_overall`

Columns:

- *externalSecondLevelPackageName* identifies the first 2 levels of the external package as described above
- *numberOfExternalCallerPackages* refers to the distinct packages that make use of the external package
- *numberOfExternalCallerTypes* refers to the distinct types that make use of the external package
- *numberOfExternalTypeCalls* includes every dependency to the types in the external package
- *numberOfExternalTypeCallsWeighted* includes every invocation or reference (sum of weights) to the types in the external package
- *allPackages* contains the total count of all analyzed packages in general
- *allTypes* contains the total count of all analyzed types in general

- *externalTypeNames* contains a list of actually utilized types of the external package

	externalSecondLevelPackageName	numberOfExternalCallerPackages	numberOfExternalCallerTypes	numberOfExternalTypeCalls	numberOfE
0	jakarta.annotation	102	684	870	
1	org.slf4j	38	74	133	
2	org.springframework	7	56	183	
3	com.fasterxml	6	16	40	
4	io.axoniq	6	45	153	
5	reactor.core	6	9	16	
6	com.google	5	9	12	
7	jakarta.persistence	5	18	42	
8	io.grpc	3	10	29	
9	org.axonframework	3	10	20	
10	org.hamcrest	3	22	44	
11	org.reactivestreams	3	6	8	
12	io.micrometer	2	12	39	
13	org.apache	2	12	42	
14	io.opentelemetry	1	5	21	
15	jakarta.validation	1	2	5	
16	javax.cache	1	2	12	
17	org.awaitility	1	1	2	
18	org.ehcache	1	3	10	
19	org.testcontainers	1	1	6	

Table 2 Chart 1a - Most called second level external packages in % by type

External package groups that are used less than 0.7% are grouped into the name "others" to get a cleaner chart with the most significant external packages and how ofter they are called in percent.

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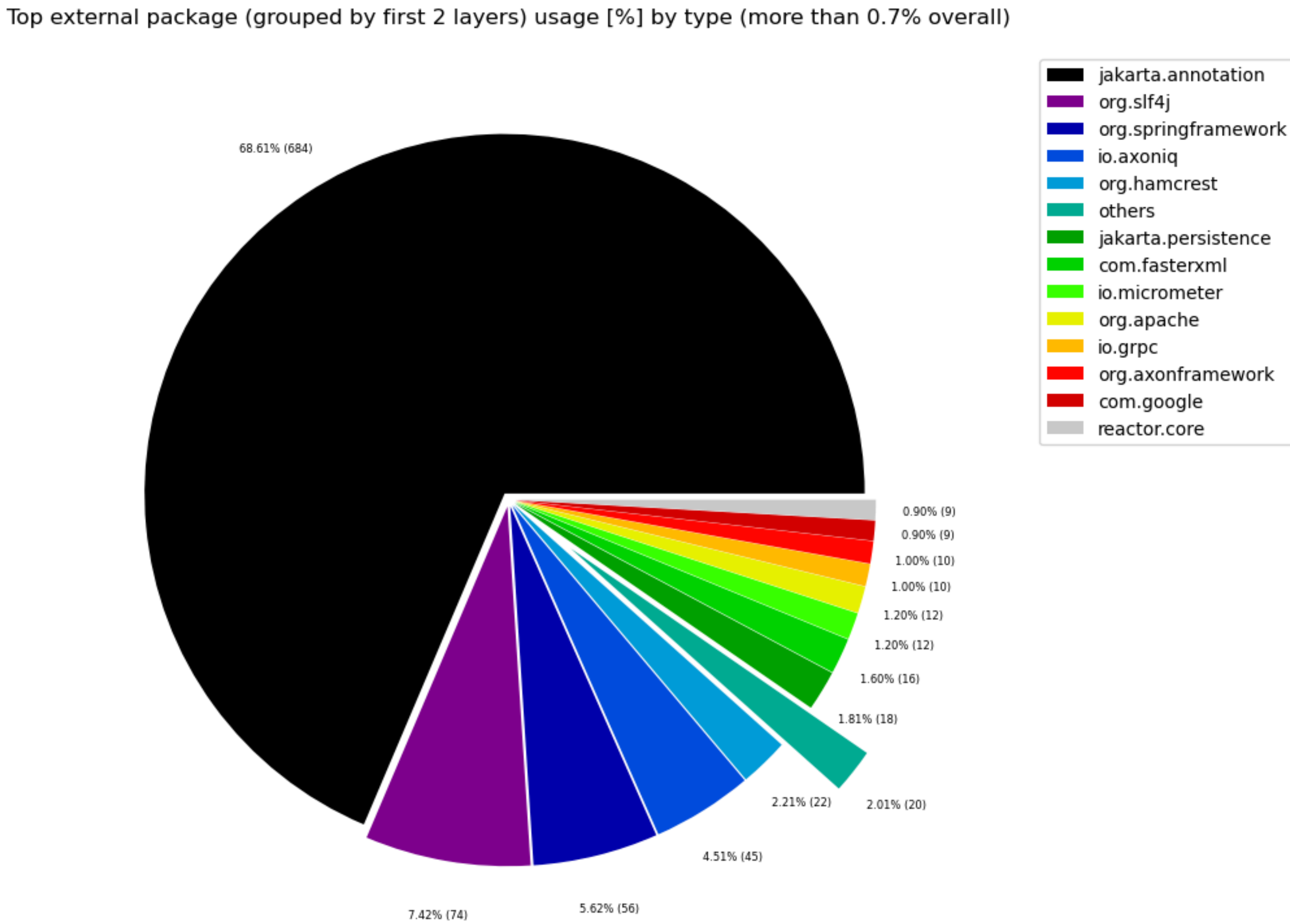

Table 2 Chart 1b - Most called second level external packages in % by type (less than 0.7% overall "others" drill-down)

Shows the lowest (less than 0.7% overall) most called external package. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.3% from that will be grouped into "others" to get a cleaner plot.

```
<Figure size 640x480 with 0 Axes>
```

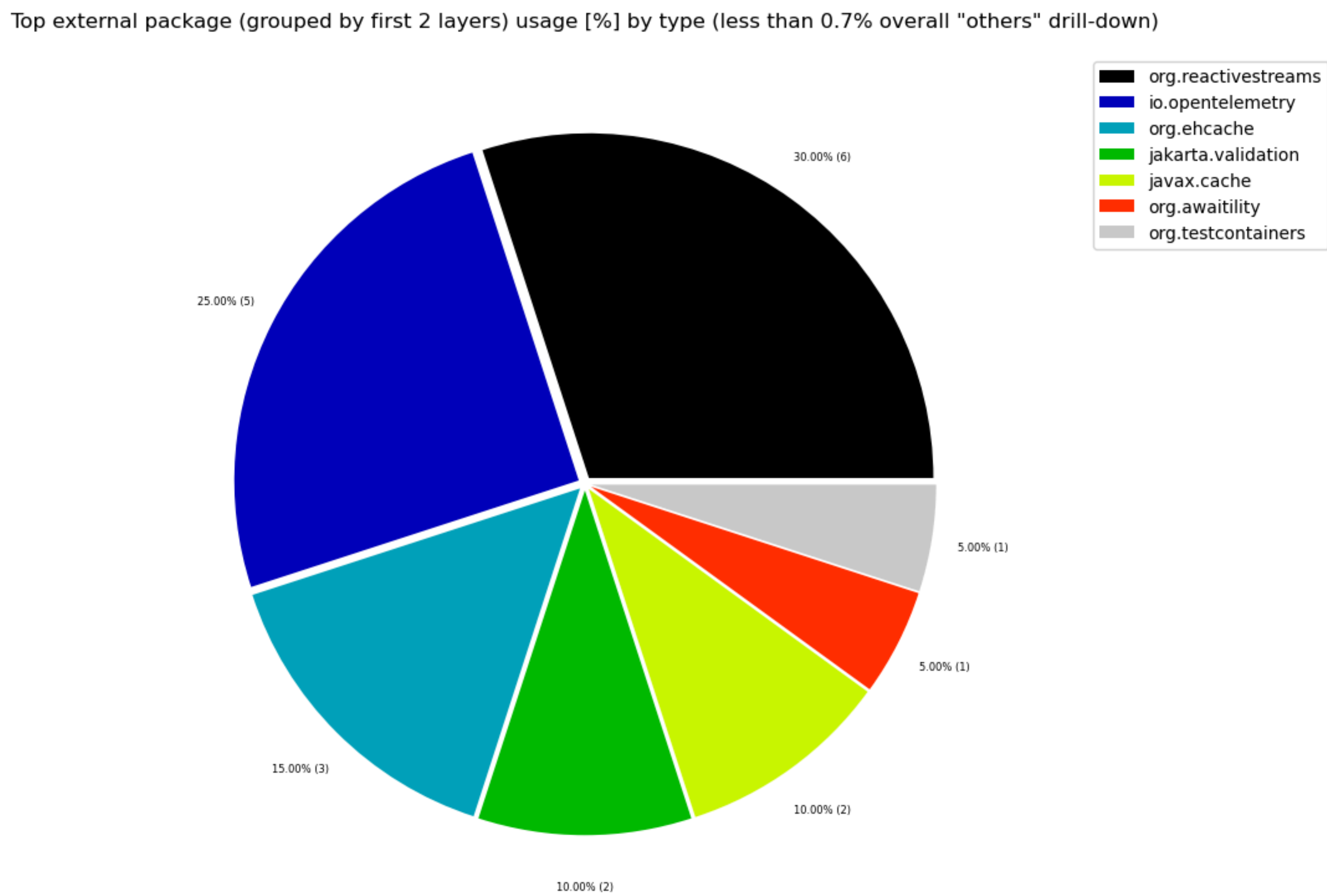

Table 2 Chart 2a - Most called second level external packages in % by package (more than 0.7% overall)

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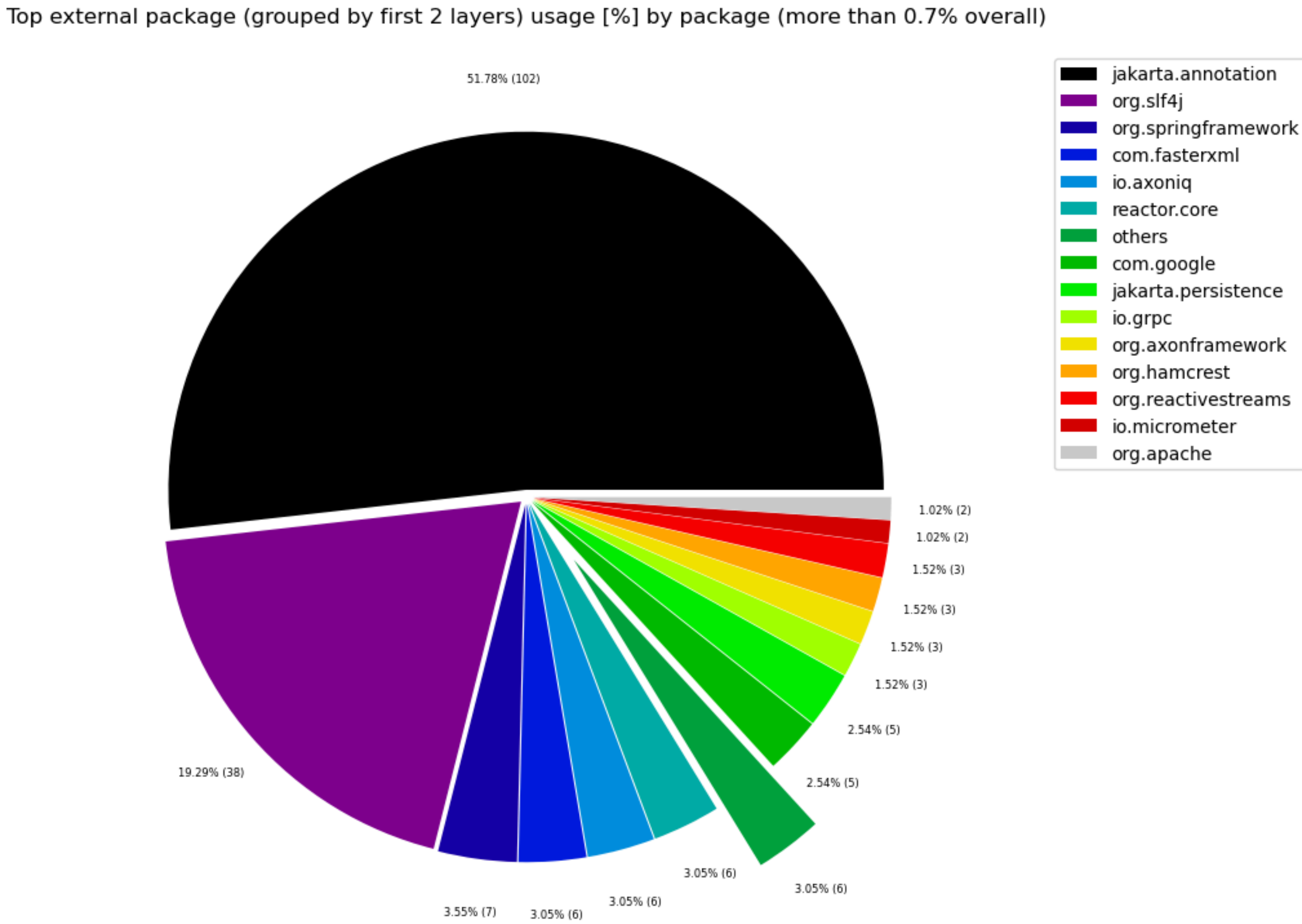

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```
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```

Top external package (grouped by first 2 layers) usage [%] by package (less than 0.7% overall "others" drill-down)

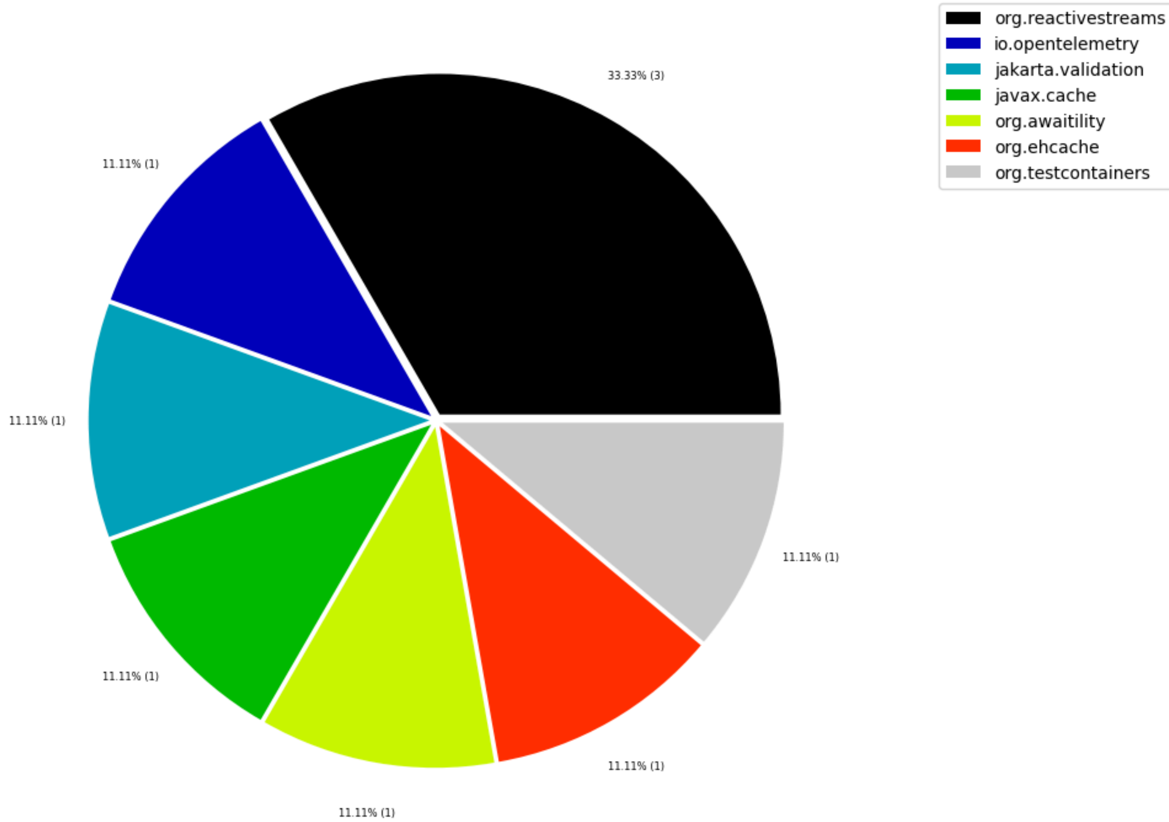

Table 3 - Top 20 most widely spread external packages

The following tables shows external packages that are used by many different artifacts with the highest number of artifacts first. External annotations are filtered out to only get those external packages that significantly add to coupling.

Statistics like minimum, maximum, average, median and standard deviation are provided for the number of packages and number of types in every artifact that uses the listed external package.

The intuition behind that is to find external package dependencies that are used in a widely spread manner. This should uncover libraries and frameworks and make it easier to distinguish them from external dependencies that are used for specific tasks. It can also be used to find external dependencies that are used sparsely regarding artifacts but are used in many different packages there. This could then be improved by applying a Hexagonal architecture.

Only the top 20 entries are shown. The whole table can be found in the following CSV report: `External_package_usage_spread`

Columns:

- *externalPackageName* identifies the external package as defined above. All other columns contain aggregated data for this external package.
- *numberOfArtifacts* contains the number of artifacts that use the external package
- *sumNumberOfPackages* contains the sum of all packages that use the external package
- *min/max/med/avg/stdNumberOfPackages* provide statistics based on the number of packages of each artifact that uses the external package
- *min/max/med/avg/stdNumberOfPackagesPercentage* provide statistics in percent (%) based on the number of packages of each artifact that uses the external package
- *min/max/med/avg/stdNumberOfTypes* provide statistics based on the number of types of each artifact that uses the external package
- *min/max/med/avg/stdNumberOfPackagesPercentage* provide statistics in percent (%) based on the number of types of each artifact that uses the external package
- *someArtifactNames* contain some of the artifacts that contain the external package for reference

	externalPackageName	numberOfArtifacts	sumNumberOfPackages	minNumberOfPackages	maxNumberOfPackages	medNumber(
0	org.slf4j	11	38	1	14	
1	jakarta.persistence	4	5	1	2	
2	com.fasterxml.jackson.databind	3	3	1	1	
3	com.fasterxml.jackson.databind.node	2	2	1	1	
4	io.axoniq.axonserver.connector	2	5	1	4	
5	io.axoniq.axonserver.connector.control	2	3	1	2	
6	org.apache.avro	2	2	1	1	
7	org.apache.avro.message	2	2	1	1	
8	IdTypeParameterResolver	1	1	1	1	
9	ScannedEntityCreator	1	1	1	1	

10 rows × 25 columns

Table 3a - Top 20 most widely spread external packages - number of internal packages

This table shows the top 20 most widely spread external packages focussing on the spread across the number of internal packages.

	externalPackageName	numberOfArtifacts	minNumberOfPackages	maxNumberOfPackages	medNumberOfPackages	av
0	org.slf4j	11	1	14	3.0	
1	jakarta.persistence	4	1	2	1.0	
2	com.fasterxml.jackson.databind	3	1	1	1.0	
3	com.fasterxml.jackson.databind.node	2	1	1	1.0	
4	io.axoniq.axonserver.connector	2	1	4	2.5	
5	io.axoniq.axonserver.connector.control	2	1	2	1.5	
6	org.apache.avro	2	1	1	1.0	
7	org.apache.avro.message	2	1	1	1.0	
8	IdTypeParameterResolver	1	1	1	1.0	
9	ScannedEntityCreator	1	1	1	1.0	
10	WrappedEventCriteriaBuilderMethod	1	1	1	1.0	
11	com.fasterxml.jackson.annotation	1	1	1	1.0	
12	com.fasterxml.jackson.core	1	1	1	1.0	
13	com.fasterxml.jackson.dataformat.cbor.databind	1	1	1	1.0	
14	com.google.gson	1	1	1	1.0	
15	com.google.protobuf	1	4	4	4.0	
16	io.axoniq.axonserver.connector.admin	1	1	1	1.0	
17	io.axoniq.axonserver.connector.command	1	1	1	1.0	
18	io.axoniq.axonserver.connector.event	1	1	1	1.0	
19	io.axoniq.axonserver.connector.impl	1	1	1	1.0	

Table 3b - Top 20 most widely spread external packages - percentage of internal packages

This table shows the top 20 most widely spread external packages focussing on the spread across the percentage of internal packages.

	externalPackageName	numberOfArtifacts	minNumberOfPackagesPercentage	maxNumberOfPackagesPercentage	medN
0	org.slf4j	11	20.000000	100.000000	
1	jakarta.persistence	4	1.754386	28.571429	
2	com.fasterxml.jackson.databind	3	6.666667	25.000000	
3	com.fasterxml.jackson.databind.node	2	6.666667	25.000000	
4	io.axoniq.axonserver.connector	2	14.285714	80.000000	
5	io.axoniq.axonserver.connector.control	2	14.285714	40.000000	
6	org.apache.avro	2	14.285714	25.000000	
7	org.apache.avro.message	2	14.285714	25.000000	
8	IdTypeParameterResolver	1	12.500000	12.500000	
9	ScannedEntityCreator	1	12.500000	12.500000	
10	WrappedEventCriteriaBuilderMethod	1	12.500000	12.500000	
11	com.fasterxml.jackson.annotation	1	1.754386	1.754386	
12	com.fasterxml.jackson.core	1	25.000000	25.000000	
13	com.fasterxml.jackson.dataformat.cbor.databind	1	14.285714	14.285714	
14	com.google.gson	1	20.000000	20.000000	
15	com.google.protobuf	1	80.000000	80.000000	
16	io.axoniq.axonserver.connector.admin	1	20.000000	20.000000	
17	io.axoniq.axonserver.connector.command	1	20.000000	20.000000	
18	io.axoniq.axonserver.connector.event	1	20.000000	20.000000	
19	io.axoniq.axonserver.connector.impl	1	20.000000	20.000000	

Table 3c - Top 20 most widely spread external packages - number of internal types

This table shows the top 20 most widely spread external packages focussing on the spread across the number of internal types.

	externalPackageName	numberOfArtifacts	minNumberOfTypes	maxNumberOfTypes	medNumberOfTypes	avgNumberOf
0	org.slf4j	11	1	30	3.0	6.7
1	jakarta.persistence	4	2	6	4.5	4.2
2	com.fasterxml.jackson.databind	3	1	5	3.0	3.0
3	com.fasterxml.jackson.databind.node	2	1	2	1.5	1.5
4	io.axoniq.axonserver.connector	2	1	15	8.0	8.0
5	io.axoniq.axonserver.connector.control	2	1	4	2.5	2.5
6	org.apache.avro	2	1	4	2.5	2.5
7	org.apache.avro.message	2	3	6	4.5	4.5
8	IdTypeParameterResolver	1	1	1	1.0	1.0
9	ScannedEntityCreator	1	1	1	1.0	1.0
10	WrappedEventCriteriaBuilderMethod	1	1	1	1.0	1.0
11	com.fasterxml.jackson.annotation	1	2	2	2.0	2.0
12	com.fasterxml.jackson.core	1	1	1	1.0	1.0
13	com.fasterxml.jackson.dataformat.cbor.databind	1	2	2	2.0	2.0
14	com.google.gson	1	1	1	1.0	1.0
15	com.google.protobuf	1	8	8	8.0	8.0
16	io.axoniq.axonserver.connector.admin	1	1	1	1.0	1.0
17	io.axoniq.axonserver.connector.command	1	1	1	1.0	1.0
18	io.axoniq.axonserver.connector.event	1	5	5	5.0	5.0
19	io.axoniq.axonserver.connector.impl	1	1	1	1.0	1.0

Table 3d - Top 20 most widely spread external packages - percentage of internal types

This table shows the top 20 most widely spread external packages focussing on the spread across the percentage of internal types.

	externalPackageName	numberOfArtifacts	minNumberOfTypesPercentage	maxNumberOfTypesPercentage	medNumberO
0	org.slf4j	11	1.369863	30.434783	
1	jakarta.persistence	4	0.350877	6.944444	
2	com.fasterxml.jackson.databind	3	0.641026	16.666667	
3	com.fasterxml.jackson.databind.node	2	0.641026	6.666667	
4	io.axoniq.axonserver.connector	2	1.388889	20.833333	
5	io.axoniq.axonserver.connector.control	2	1.388889	5.555556	
6	org.apache.avro	2	1.388889	13.333333	
7	org.apache.avro.message	2	4.166667	20.000000	
8	IdTypeParameterResolver	1	0.961538	0.961538	
9	ScannedEntityCreator	1	0.961538	0.961538	
10	WrappedEventCriteriaBuilderMethod	1	0.961538	0.961538	
11	com.fasterxml.jackson.annotation	1	0.350877	0.350877	
12	com.fasterxml.jackson.core	1	3.333333	3.333333	
13	com.fasterxml.jackson.dataformat.cbor.databind	1	2.777778	2.777778	
14	com.google.gson	1	1.369863	1.369863	
15	com.google.protobuf	1	11.111111	11.111111	
16	io.axoniq.axonserver.connector.admin	1	1.388889	1.388889	
17	io.axoniq.axonserver.connector.command	1	1.388889	1.388889	
18	io.axoniq.axonserver.connector.event	1	6.944444	6.944444	
19	io.axoniq.axonserver.connector.impl	1	1.388889	1.388889	

Table 3 Chart 1a - Most widely spread external packages in % by types (more than 0.5% overall)

External packages that are used less than 0.5% are grouped into the name "others" to get a cleaner chart with the most significant external packages.

```
<Figure size 640x480 with 0 Axes>
```

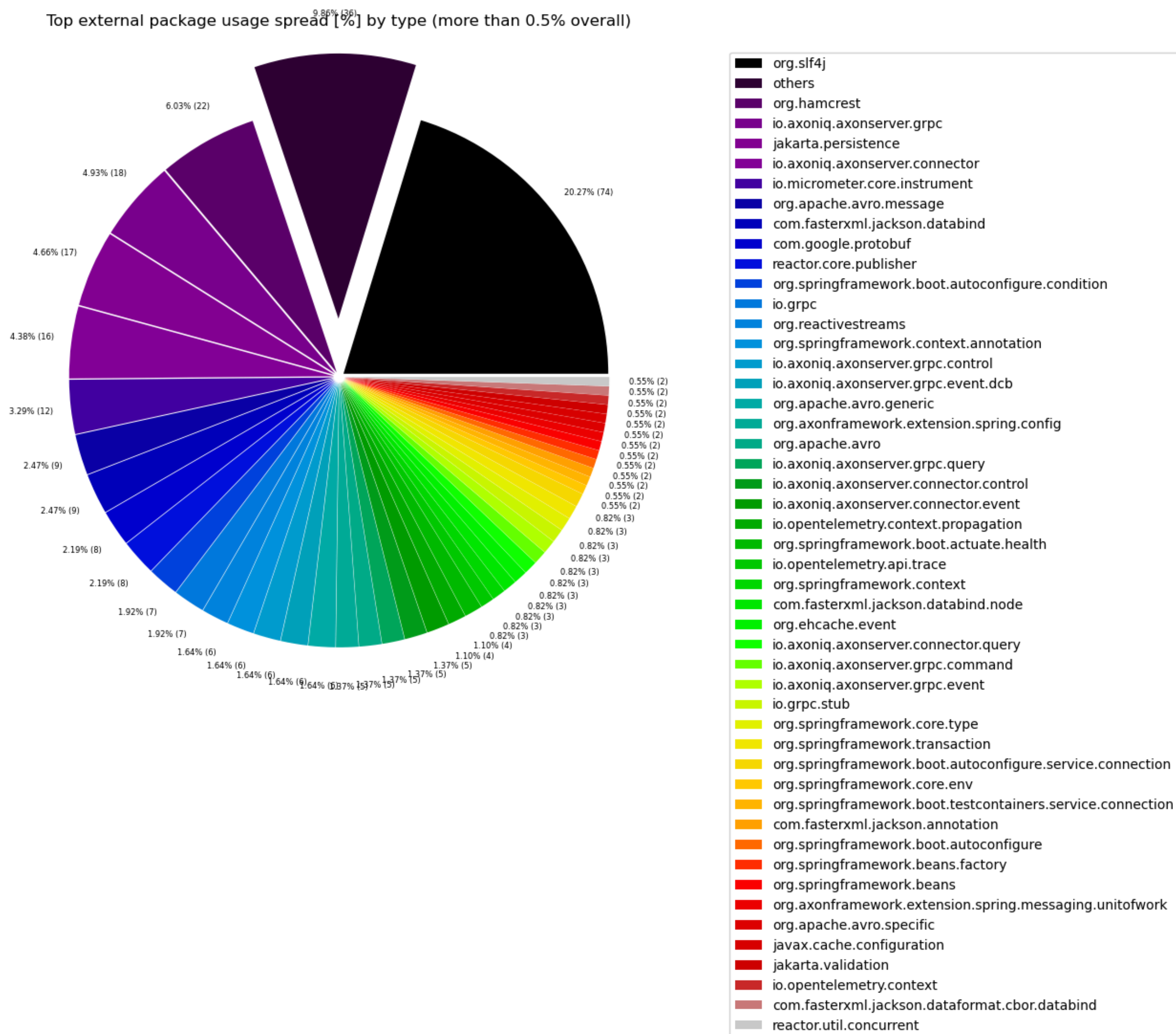

Table 3 Chart 1b - Most widely spread external packages in % by types (less than 0.5% overall "others" drill-down)

Shows the lowest (less than 0.5% overall) most spread external packages. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.3% from that will be grouped into "others" to get a cleaner plot.

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```

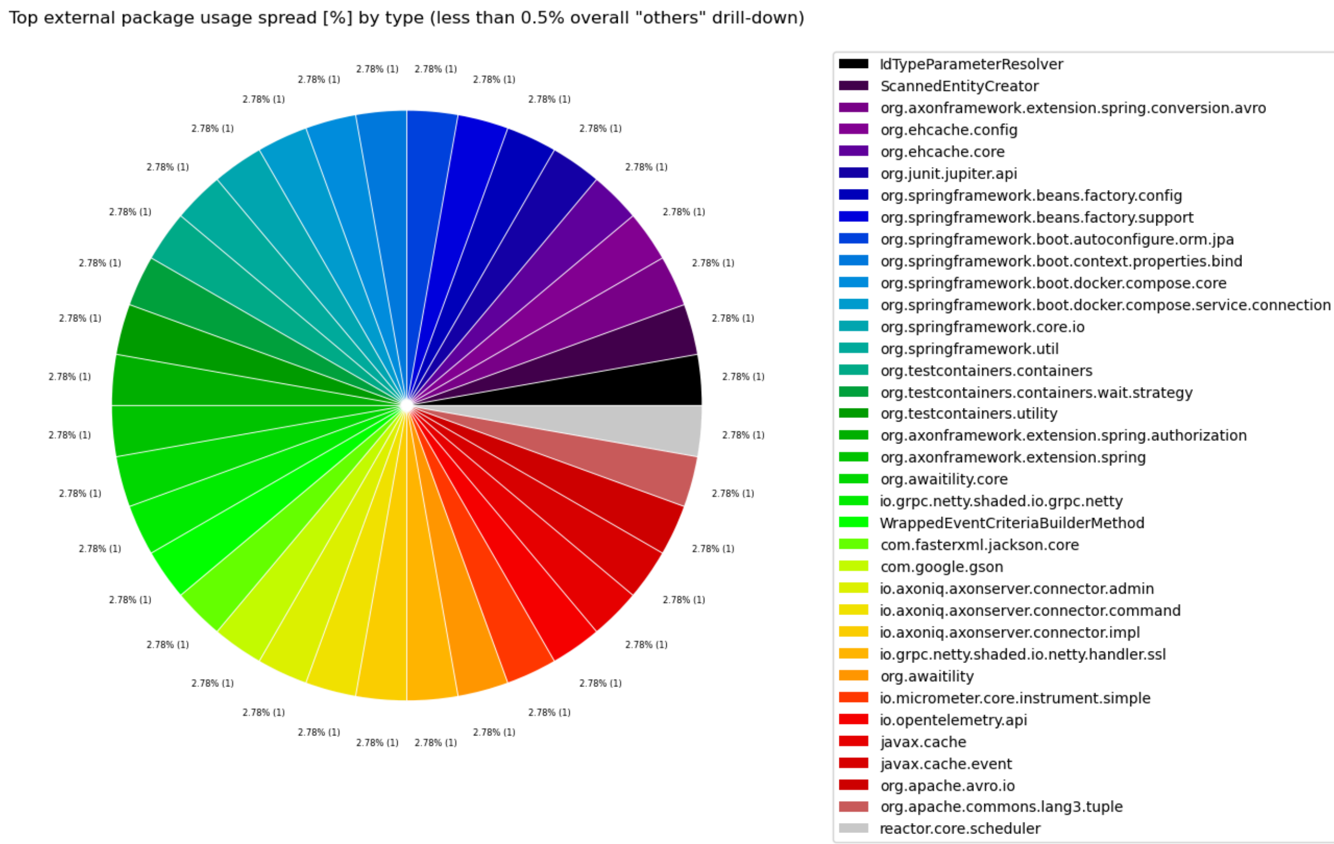

Table 3 Chart 2a - Most widely spread external packages in % by packages (more than 0.5% overall)

External packages that are used less than 0.5% are grouped into the name "others" to get a cleaner chart with the most significant external packages.

```
<Figure size 640x480 with 0 Axes>
```

Top external package usage spread [%] by package (more than 0.5% overall)

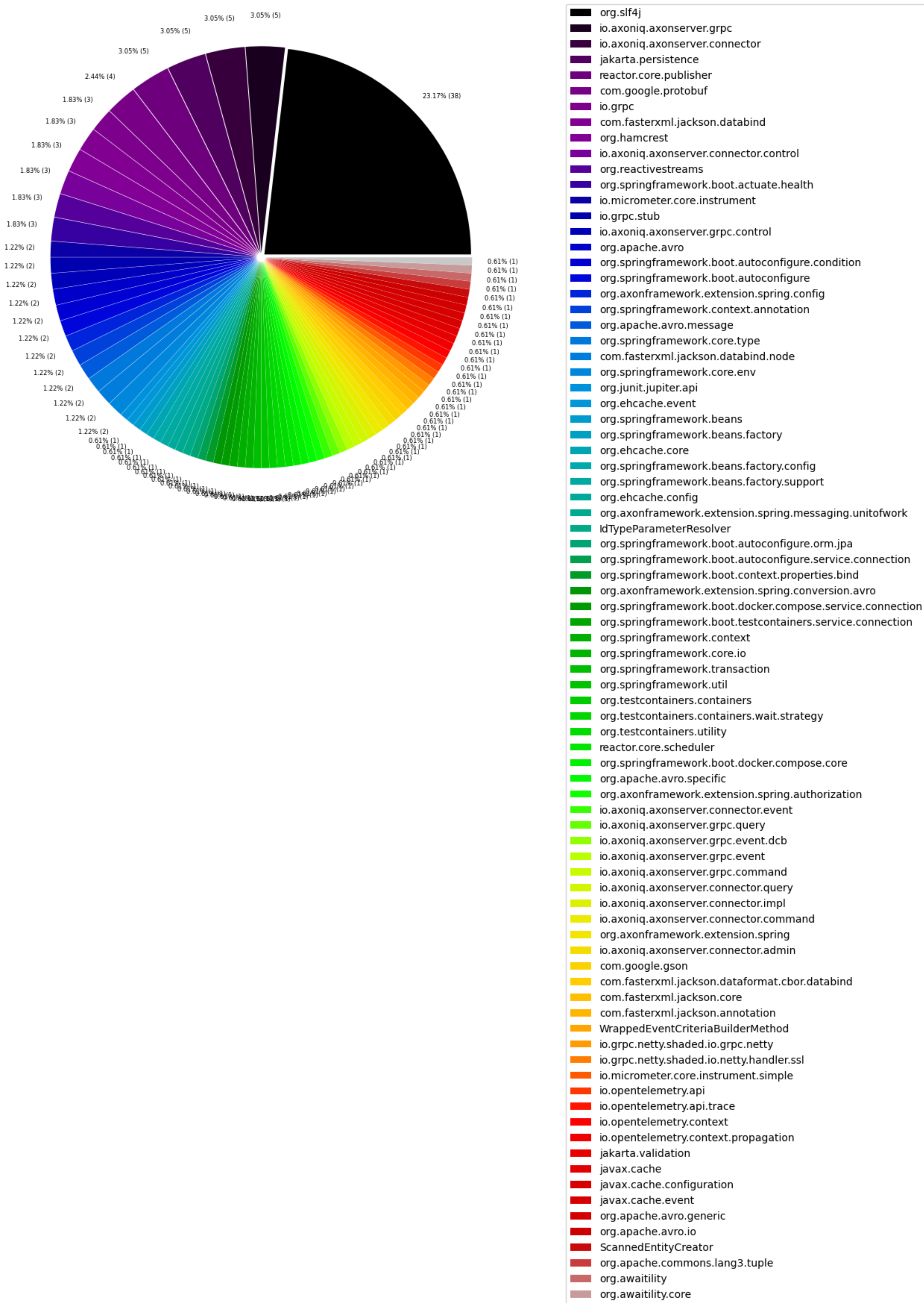

Table 3 Chart 2b - Most widely spread external packages in % by packages (less than 0.5% overall "others" drill-down)

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```
No data to plot for title 'Top external package usage spread [%] by type (less than
0.7% overall "others" drill-down)'.
```

Table 4 - Top 20 most widely spread external packages grouped by their first 2 layers

This table shows external packages grouped by their first 2 layers that are used by many different artifacts with the highest number of artifacts first. External annotations are filtered out to only get those external packages that significantly add to coupling.

Statistics like minimum, maximum, average, median and standard deviation are provided for the number of packages and number of types in every artifact that uses the listed external package.

The intuition behind that is to find external package dependencies that are used in a widely spread manner. This should uncover libraries and frameworks and make it easier to distinguish them from external dependencies that are used for specific tasks. It can also be used to find external dependencies that are used sparsely regarding artifacts but are used in many different packages there. This could then be improved by applying a Hexagonal architecture.

Only the top 20 entries are shown. The whole table can be found in the following CSV report: `External_package_usage_spread`

Columns:

- *externalPackageName* identifies the external package as defined above. All other columns contain aggregated data for this external package.
- *numberOfArtifacts* contains the number of artifacts that use the external package
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- *min/max/med/avg/stdNumberOfPackages* provide statistics based on the number of packages of each artifact that uses the external package
- *min/max/med/avg/stdNumberOfPackagesPercentage* provide statistics in percent (%) based on the number of packages of each artifact that uses the external package
- *min/max/med/avg/stdNumberOfTypes* provide statistics based on the number of types of each artifact that uses the external package

- *min/max/med/avg/stdNumberOfPackagesPercentage* provide statistics in percent (%) based on the number of types of each artifact that uses the external package
- *someArtifactNames* contain some of the artifacts that contain the external package for reference

	externalSecondLevelPackageName	numberOfArtifacts	sumNumberOfPackages	minNumberOfPackages	maxNumberOfPackages	medNumbe
0	org.slf4j	11	38	1	14	
1	com.fasterxml	4	4	1	1	
2	jakarta.persistence	4	5	1	2	
3	com.google	2	5	1	4	
4	io.axoniq	2	6	1	5	
5	org.apache	2	2	1	1	
6	reactor.core	2	6	1	5	
7	IdTypeParameterResolver	1	1	1	1	
8	ScannedEntityCreator	1	1	1	1	
9	WrappedEventCriteriaBuilderMethod	1	1	1	1	
10	io.grpc	1	3	3	3	
11	io.micrometer	1	2	2	2	
12	io.opentelemetry	1	1	1	1	
13	jakarta.validation	1	1	1	1	
14	javax.cache	1	1	1	1	
15	org.awaitility	1	1	1	1	
16	org.axonframework	1	3	3	3	
17	org.ehcache	1	1	1	1	
18	org.hamcrest	1	3	3	3	
19	org.junit	1	1	1	1	

20 rows × 25 columns

Table 4 Chart 1a - Most widely spread second level external packages in % by type (more than 0.5% overall)

External package groups that are used less than 0.5% are grouped into the name "others" to get a cleaner chart with the most significant external packages and how ofter they are called in percent.

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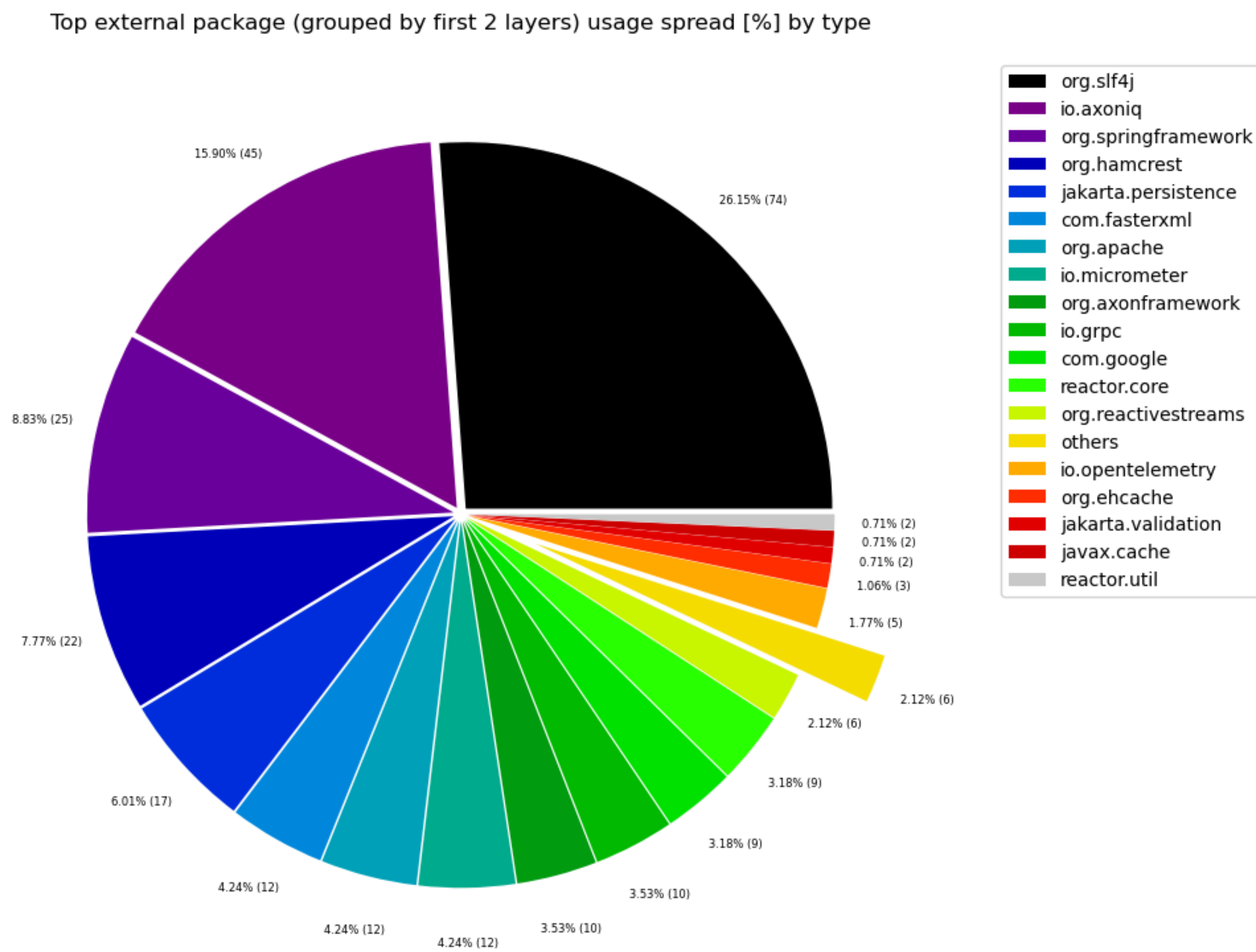

Table 4 Chart 1b - Most widely spread second level external packages in % by type (less than 0.5% overall "others" drill-down)

External packages that are used less than 0.5% are grouped into the name "others" to get a cleaner chart with the most significant external packages.

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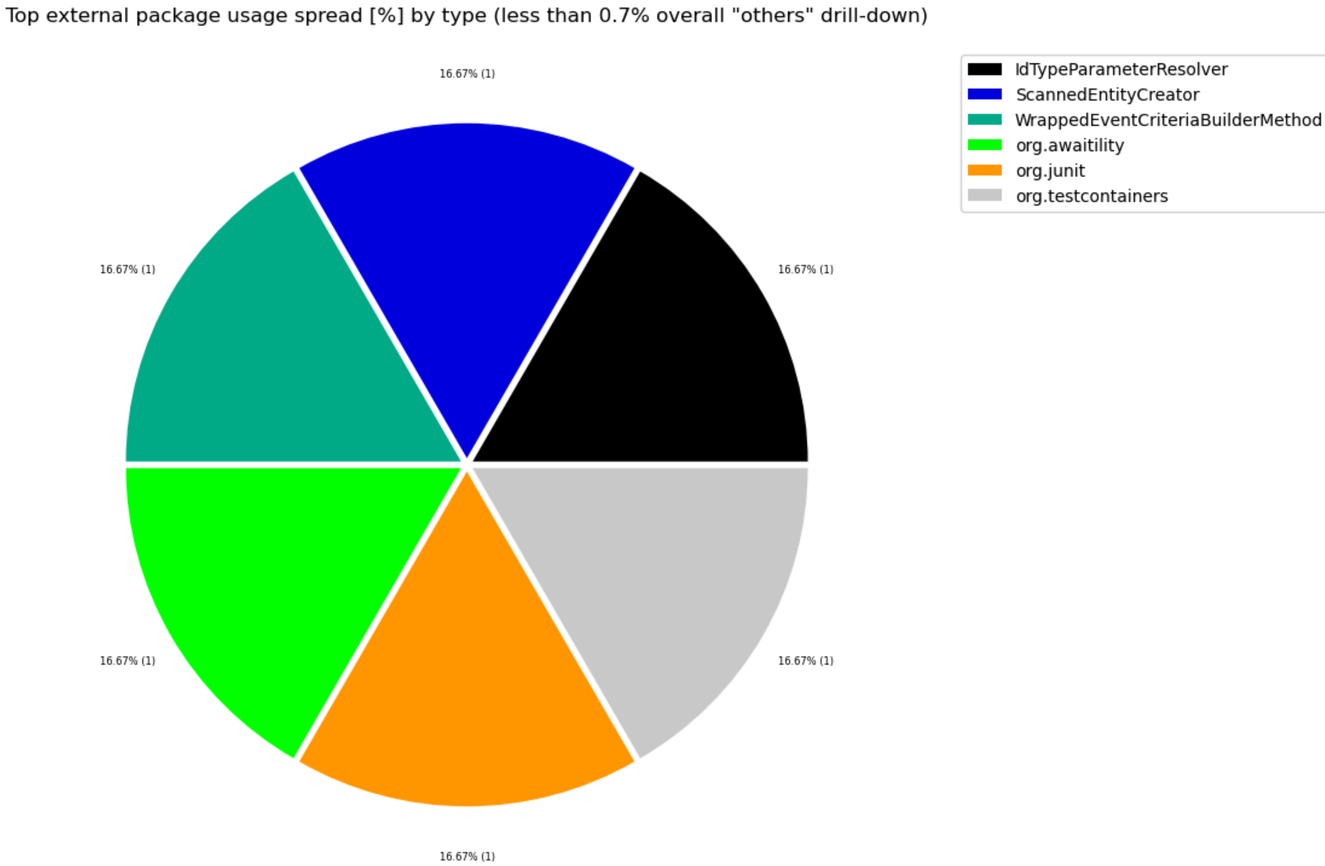


Table 4 Chart 2a - Most widely spread second level external packages in % by package (more than 0.5% overall)

External package groups that are used less than 0.5% are grouped into the name "others" to get a cleaner chart with the most significant external packages and how ofter they are called in percent.

```
<Figure size 640x480 with 0 Axes>
```

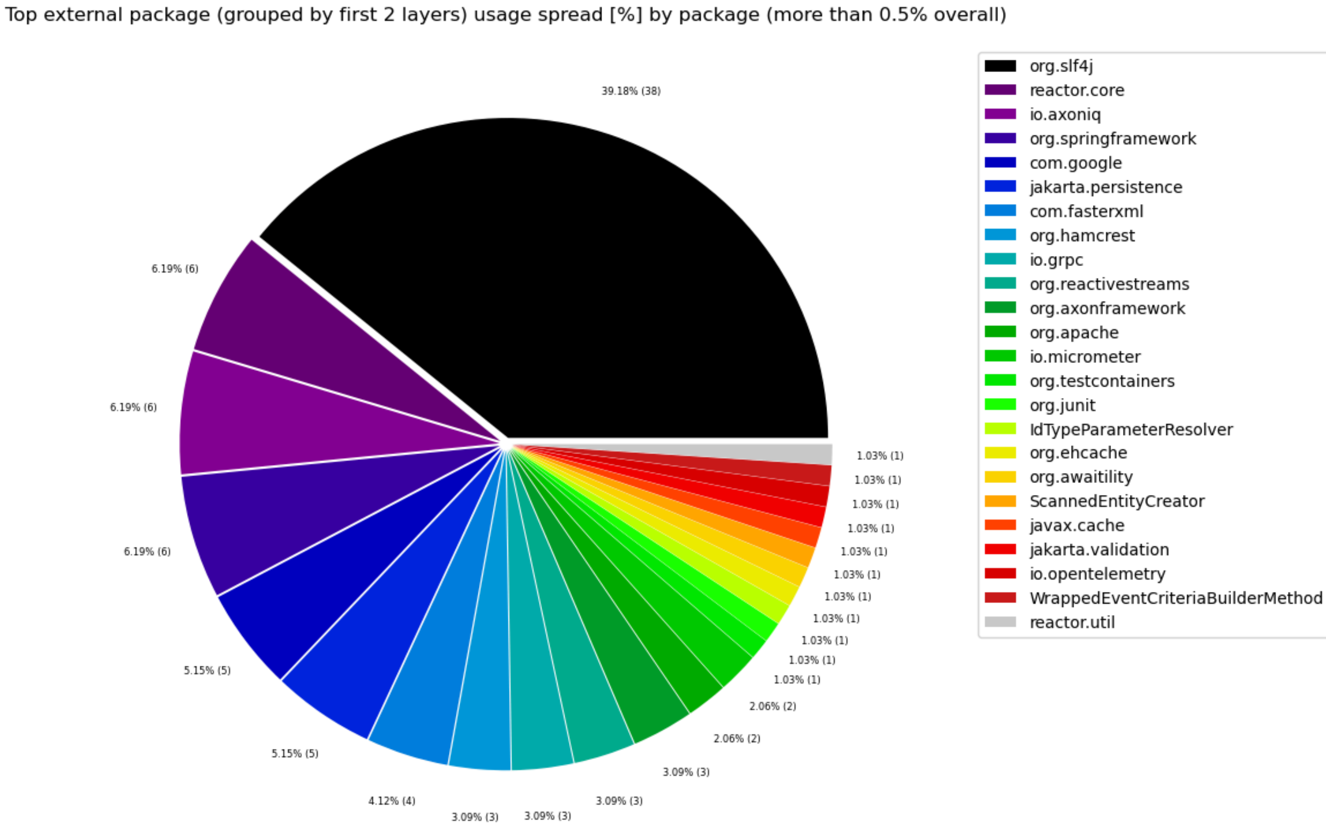


Table 4 Chart 2b - Most widely spread second level external packages in % by package (less than 0.5% overall "others" drill-down)

External packages that are used less than 0.5% are grouped into the name "others" to get a cleaner chart with the most significant external packages.

```
No data to plot for title 'Top external package (less than 0.7% overall "others" dril
l-down)'.
```

Table 5 - Top 20 least used external packages overall

This table identifies external packages that aren't used very often. This could help to find libraries that aren't actually needed or maybe easily replaceable. Some of them might be used sparsely on purpose for example as an adapter to an external library that is actually important. Thus, decisions need to be made on a case-by-case basis.

Only the last 20 entries are shown. The whole table can be found in the following CSV report: `External_package_usage_overall`

Columns:

- *externalPackageName* identifies the external package as described above
- *numberOfExternalTypeCalls* includes every invocation or reference to the types in the external package

	externalPackageName	numberOfExternalTypeCalls
0	reactor.core.scheduler	2
1	org.springframework.boot.docker.compose.core	2
2	org.springframework.boot.docker.compose.servic...	2
3	org.testcontainers.utility	2
4	org.testcontainers.containers	2
5	org.testcontainers.containers.wait.strategy	2
6	javax.cache.configuration	3
7	org.axonframework.extension.spring.conversion....	3
8	org.springframework.boot.testcontainers.servic...	4
9	io.grpc.stub	4
10	com.google.gson	4
11	org.apache.avro.specific	4
12	org.springframework.beans.factory	4
13	com.fasterxml.jackson.databind.node	5
14	org.springframework.context	5
15	org.springframework.boot.context.properties.bind	5
16	jakarta.validation	5
17	io.axoniq.axonserver.grpc.event	5
18	io.axoniq.axonserver.connector.query	6
19	org.apache.avro.generic	7

Table 6 - External usage per artifact sorted by highest external type rate descending

The following table shows the most used external packages separately for each artifact including external annotations. The results are sorted by the artifacts with the highest external type usage rate descending.

The intention of this table is to find artifacts that use a lot of external dependencies in relation to their size and get all the external packages and their usage.

Only the last 40 entries are shown. The whole table can be found in the following CSV report: `External_package_usage_per_artifact_sorted`

Columns:

- *artifactName* is used to group the the external package usage per artifact for a more detailed analysis.
- *externalPackageName* identifies the external package as described above
- *numberOfExternalTypeCaller* refers to the distinct types that make use of the external package
- *numberOfExternalTypeCalls* includes every invocation or reference to the types in the external package
- *numberOfTypesInArtifact* represents the total count of all analyzed types for the artifact
- *numberOfExternalTypesInArtifact* is the number of all external types that are used by the artifact

- *numberOfExternalPackagesInArtifact* is the number of all external packages that are used by the artifact
- *externalTypeRate* is the numberOfExternalTypesInArtifact / numberOfTypesInArtifact * 100
- *externalTypeNames* contains a list of actually utilized types of the external package

	artifactName	externalPackageName	numberOfExternalTypeCaller	numberOfExternalTypeCalls	numberOfTypesInArtifa
0	axon-tracing-opentelemetry-5.0.2	io.opentelemetry.api.trace	9	46	
1	axon-tracing-opentelemetry-5.0.2	io.opentelemetry.context.propagation	9	15	
2	axon-tracing-opentelemetry-5.0.2	jakarta.annotation	3	8	
3	axon-tracing-opentelemetry-5.0.2	io.opentelemetry.context	2	7	
4	axon-tracing-opentelemetry-5.0.2	org.slf4j	2	7	
5	axon-tracing-opentelemetry-5.0.2	io.opentelemetry.api	1	2	
6	axon-server-connector-5.0.2	io.axoniq.axonserver.grpc	40	271	
7	axon-server-connector-5.0.2	jakarta.annotation	31	181	
8	axon-server-connector-5.0.2	io.axoniq.axonserver.grpc.event.dcb	28	104	
9	axon-server-connector-5.0.2	io.grpc	23	51	
10	axon-server-connector-5.0.2	io.axoniq.axonserver.connector	22	123	
11	axon-server-connector-5.0.2	org.slf4j	19	70	
12	axon-server-connector-5.0.2	io.axoniq.axonserver.grpc.query	14	98	
13	axon-server-connector-5.0.2	io.axoniq.axonserver.grpc.control	13	86	
14	axon-server-connector-5.0.2	com.google.protobuf	8	29	
15	axon-server-connector-5.0.2	io.axoniq.axonserver.connector.event	8	38	
16	axon-server-connector-5.0.2	io.axoniq.axonserver.grpc.command	8	63	
17	axon-server-connector-5.0.2	io.axoniq.axonserver.connector.query	6	19	
18	axon-server-connector-5.0.2	io.axoniq.axonserver.grpc.event	5	31	
19	axon-server-connector-5.0.2	io.axoniq.axonserver.connector.control	4	5	
20	axon-server-connector-5.0.2	io.grpc.stub	4	11	
21	axon-server-connector-5.0.2	reactor.core.scheduler	2	2	
22	axon-server-connector-5.0.2	io.axoniq.axonserver.connector.admin	1	3	
23	axon-server-connector-5.0.2	io.axoniq.axonserver.connector.command	1	3	
24	axon-server-connector-5.0.2	io.axoniq.axonserver.connector.impl	1	3	
25	axon-server-connector-5.0.2	io.grpc.netty.shaded.io.grpc.netty	1	1	
26	axon-server-connector-	io.grpc.netty.shaded.io.netty.handler.ssl	1	2	

	artifactName	externalPackageName	numberOfExternalTypeCaller	numberOfExternalTypeCalls	numberOfTypesInArtifa
	5.0.2				
27	axon-server-connector-5.0.2	javax.annotation	1	1	
28	axon-server-connector-5.0.2	org.springframework.boot.context.properties	1	2	
29	axon-spring-boot-autoconfigure-5.0.2	org.springframework.boot.autoconfigure.condition	45	75	
30	axon-spring-boot-autoconfigure-5.0.2	org.springframework.context.annotation	36	65	
31	axon-spring-boot-autoconfigure-5.0.2	org.springframework.boot.autoconfigure	30	31	
32	axon-spring-boot-autoconfigure-5.0.2	org.springframework.boot.context.properties	20	26	
33	axon-spring-boot-autoconfigure-5.0.2	jakarta.annotation	15	30	
34	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension.spring.config	13	42	
35	axon-spring-boot-autoconfigure-5.0.2	org.springframework.boot.actuate.health	7	24	
36	axon-spring-boot-autoconfigure-5.0.2	jakarta.persistence	6	12	
37	axon-spring-boot-autoconfigure-5.0.2	org.springframework.boot.context.properties.bind	5	11	
38	axon-spring-boot-autoconfigure-5.0.2	org.springframework.context	5	9	
39	axon-spring-boot-autoconfigure-5.0.2	org.apache.avro.message	4	10	

Table 7 - Artifacts and their external packages

The following table shows the artifacts with the highest external dependency usage broken down by each external package including external annotations. The results are sorted by the artifacts with the highest external package usage rate descending.

The intention of this table is to find artifacts that use a lot of external dependencies and show in detail which external packages are used by them and how many internal packages.

Only the last 30 entries are shown. The whole table can be found in the following CSV report: `External_package_usage_per_artifact_and_external_package`

Columns:

- *artifactName* is the name of the artifact with external dependencies (first grouping column)
- *artifactPackages* is the number of packages in the artifact

- *artifactTypes* is the number of types in the artifact
- *artifactExternalPackages* is the number of external packages used by the artifact
- *artifactExternalCallingPackages* is the number of packages that use external packages in the artifact
- *artifactExternalCallingPackagesRate* is artifactExternalCallingPackages / artifactPackages * 100%
- *externalPackageName* the name of the external package (second grouping column)
- *numberOfPackages* is the number of internal packages of the artifact that use the external packages
- *numberOfTypes* is the number of internal types of the artifact that use the external packages
- *packagesCallingExternalRate* is numberOfPackages / artifactPackages * 100%
- *typesCallingExternalRate* is numberOfTypes / artifactTypes * 100%
- *nameOfPackages* names of the internal packages that use the external package in the artifact
- *someTypeNames* some (10) names of the internal types that use the external package in the artifact

	artifactName	artifactPackages	artifactTypes	artifactExternalPackages	artifactExternalCallingPackages	artifactExternalCallingPackagesRat
0	axon-conversion-5.0.2	4	30	11	4	100.
1	axon-conversion-5.0.2	4	30	11	4	100.
2	axon-conversion-5.0.2	4	30	11	4	100.
3	axon-conversion-5.0.2	4	30	11	4	100.
4	axon-conversion-5.0.2	4	30	11	4	100.
5	axon-conversion-5.0.2	4	30	11	4	100.
6	axon-conversion-5.0.2	4	30	11	4	100.
7	axon-conversion-5.0.2	4	30	11	4	100.
8	axon-conversion-5.0.2	4	30	11	4	100.
9	axon-conversion-5.0.2	4	30	11	4	100.
10	axon-conversion-5.0.2	4	30	11	4	100.
11	axon-eventsourcing-5.0.2	8	104	6	8	100.
12	axon-eventsourcing-5.0.2	8	104	6	8	100.
13	axon-eventsourcing-5.0.2	8	104	6	8	100.
14	axon-eventsourcing-5.0.2	8	104	6	8	100.
15	axon-eventsourcing-5.0.2	8	104	6	8	100.
16	axon-eventsourcing-5.0.2	8	104	6	8	100.
17	axon-metrics-micrometer-5.0.2	2	13	4	2	100.
18	axon-metrics-micrometer-5.0.2	2	13	4	2	100.
19	axon-metrics-micrometer-5.0.2	2	13	4	2	100.
20	axon-metrics-micrometer-5.0.2	2	13	4	2	100.
21	axon-modelling-5.0.2	7	93	2	7	100.
22	axon-modelling-5.0.2	7	93	2	7	100.
23	axon-server-connector-5.0.2	5	72	23	5	100.
24	axon-server-connector-5.0.2	5	72	23	5	100.
25	axon-server-connector-5.0.2	5	72	23	5	100.
26	axon-server-connector-	5	72	23	5	100.

	artifactName	artifactPackages	artifactTypes	artifactExternalPackages	artifactExternalCallingPackages	artifactExternalCallingPackagesRat
	5.0.2					
27	axon-server-connector-5.0.2	5	72	23	5	100.
28	axon-server-connector-5.0.2	5	72	23	5	100.
29	axon-server-connector-5.0.2	5	72	23	5	100.

Table 7a - Artifacts and their external packages (first 2 levels)

The following table groups the external packages by their first two levels. For example `javax.xml.namespace` and `javax.xml.stream` will be grouped together to `javax.xml`.

	artifactName	artifactPackages	artifactTypes	artifactExternalPackagesFirst2Levels	artifactExternalCallingPackages	artifactExternalCalling
0	axon-conversion-5.0.2	4	30	4	4	
1	axon-conversion-5.0.2	4	30	4	4	
2	axon-conversion-5.0.2	4	30	4	4	
3	axon-conversion-5.0.2	4	30	4	4	
4	axon-eventsourcing-5.0.2	8	104	6	8	
5	axon-eventsourcing-5.0.2	8	104	6	8	
6	axon-eventsourcing-5.0.2	8	104	6	8	
7	axon-eventsourcing-5.0.2	8	104	6	8	
8	axon-eventsourcing-5.0.2	8	104	6	8	
9	axon-eventsourcing-5.0.2	8	104	6	8	
10	axon-metrics-micrometer-5.0.2	2	13	3	2	
11	axon-metrics-micrometer-5.0.2	2	13	3	2	
12	axon-metrics-micrometer-5.0.2	2	13	3	2	
13	axon-modelling-5.0.2	7	93	2	7	
14	axon-modelling-5.0.2	7	93	2	7	
15	axon-server-connector-5.0.2	5	72	8	5	
16	axon-server-connector-5.0.2	5	72	8	5	
17	axon-server-connector-5.0.2	5	72	8	5	
18	axon-server-connector-5.0.2	5	72	8	5	
19	axon-server-connector-5.0.2	5	72	8	5	
20	axon-server-connector-5.0.2	5	72	8	5	
21	axon-server-connector-5.0.2	5	72	8	5	
22	axon-server-connector-5.0.2	5	72	8	5	
23	axon-spring-boot-autoconfigure-5.0.2	7	72	8	7	
24	axon-spring-boot-autoconfigure-5.0.2	7	72	8	7	
25	axon-spring-boot-	7	72	8	7	

	artifactName	artifactPackages	artifactTypes	artifactExternalPackagesFirst2Levels	artifactExternalCallingPackages	artifactExternalCalling
	autoconfigure-5.0.2					
26	axon-spring-boot-autoconfigure-5.0.2	7	72	8	7	
27	axon-spring-boot-autoconfigure-5.0.2	7	72	8	7	
28	axon-spring-boot-autoconfigure-5.0.2	7	72	8	7	
29	axon-spring-boot-autoconfigure-5.0.2	7	72	8	7	

Table 7b - Top 15 external dependency using artifacts as columns with their external packages

The following table uses pivot to show the artifacts in columns, the external dependencies in rows and the number of internal packages as values.

artifactName	axon-messaging-5.0.2	axon-spring-boot-autoconfigure-5.0.2	axon-server-connector-5.0.2	axon-common-5.0.2	axon-conversion-5.0.2	axon-test-5.0.2	axon-eventsourcing-5.0.2	axon-modelling-5.0.2
externalPackageName								
IdTypeParameterResolver	0	0	0	0	0	0	1	0
ScannedEntityCreator	0	0	0	0	0	0	1	0
WrappedEventCriteriaBuilderMethod	0	0	0	0	0	0	1	0
com.fasterxml.jackson.annotation	3	0	0	0	0	0	0	0
com.fasterxml.jackson.core	0	0	0	0	1	0	0	0
...	...	...	...	...	...	...	...	...
org.testcontainers.containers.wait.strategy	0	0	0	0	0	1	0	0
org.testcontainers.utility	0	0	0	0	0	1	0	0
reactor.core.publisher	5	0	0	0	0	0	0	0
reactor.core.scheduler	0	0	1	0	0	0	0	0
reactor.util.concurrent	1	0	0	0	0	0	0	0

90 rows × 11 columns

Table 7c - Top 15 external dependency using artifacts as columns with their external packages (first 2 levels)

The following table uses pivot to show the artifacts in columns, the external package name grouped by its first two levels in rows and the number of internal packages as values. For example `javax.xml.namespace` and `javax.xml.stream` will be grouped together to `javax.xml`.

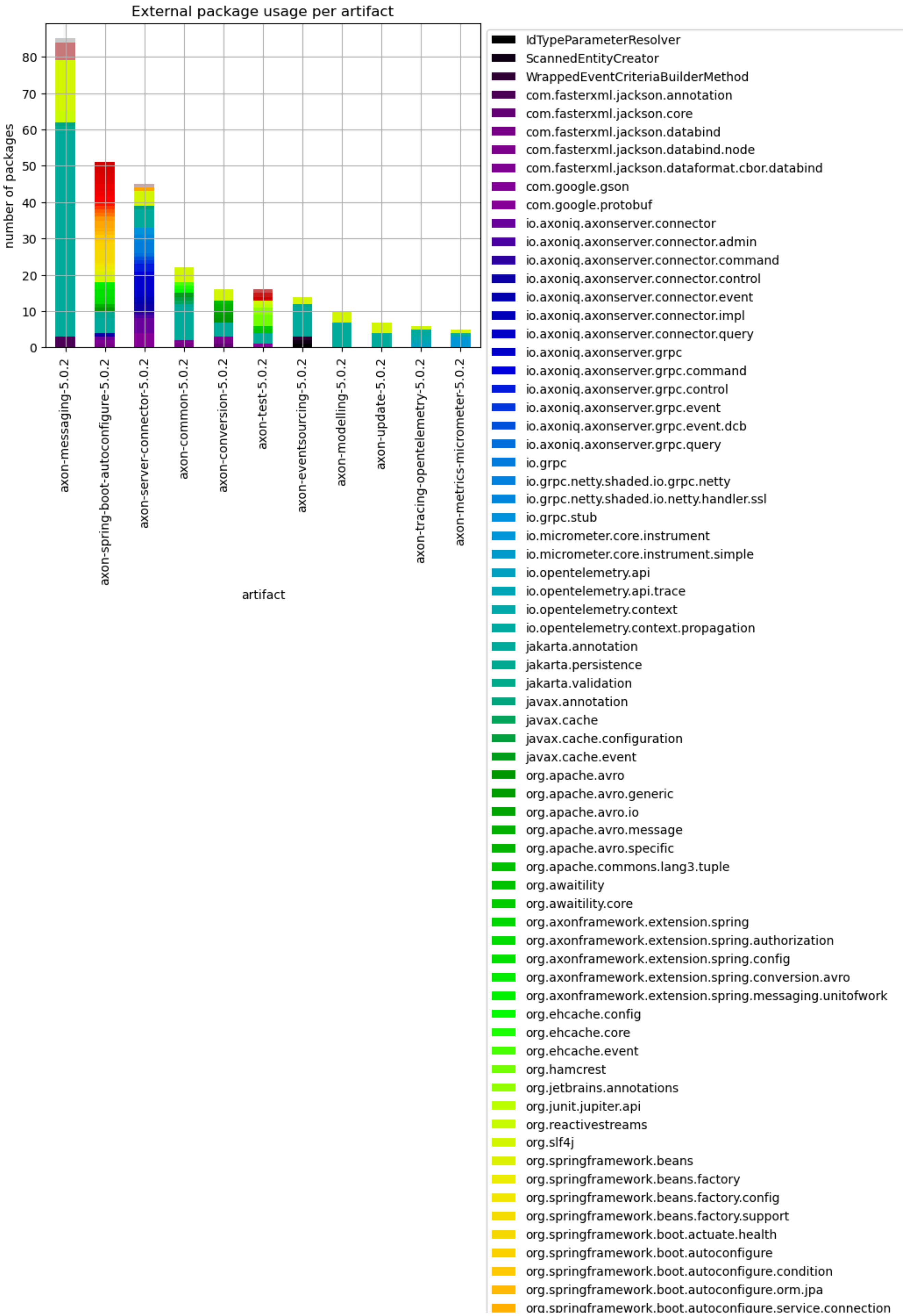
artifactName	axon-messaging-5.0.2	axon-server-connector-5.0.2	axon-spring-boot-autoconfigure-5.0.2	axon-common-5.0.2	axon-eventsourcing-5.0.2	axon-test-5.0.2	axon-modelling-5.0.2	axon-conversion-5.0.2	axon update 5.0.
externalPackageNameFirst2Levels									
IdTypeParameterResolver	0	0	0	0	1	0	0	0	
ScannedEntityCreator	0	0	0	0	1	0	0	0	
WrappedEventCriteriaBuilderMethod	0	0	0	0	1	0	0	0	
com.fasterxml	3	0	1	1	0	0	0	1	
com.google	0	4	0	0	0	1	0	0	
io.axoniq	0	5	1	0	0	0	0	0	
io.grpc	0	3	0	0	0	0	0	0	
io.micrometer	0	0	0	0	0	0	0	0	
io.opentelemetry	0	0	0	0	0	0	0	0	
jakarta.annotation	56	5	4	9	8	3	7	4	
jakarta.persistence	1	0	2	1	1	0	0	0	
jakarta.validation	1	0	0	0	0	0	0	0	
javax.annotation	1	1	0	0	0	0	0	0	
javax.cache	0	0	0	1	0	0	0	0	
org.apache	0	0	1	0	0	0	0	1	
org.awaitility	0	0	0	0	0	1	0	0	
org.axonframework	0	0	3	0	0	0	0	0	
org.ehcache	0	0	0	1	0	0	0	0	
org.hamcrest	0	0	0	0	0	3	0	0	
org.jetbrains	0	0	0	0	0	2	0	0	
org.junit	0	0	0	0	0	1	0	0	
org.reactivestreams	3	0	0	0	0	0	0	0	
org.slf4j	14	4	2	4	2	1	3	3	
org.springframework	0	1	6	0	0	0	0	0	
org.testcontainers	0	0	0	0	0	1	0	0	
reactor.core	5	1	0	0	0	0	0	0	
reactor.util	1	0	0	0	0	0	0	0	

Table 7 Chart 1 - Top 15 external dependency using artifacts and their external packages stacked

The following chart shows the top 15 external package using artifacts and breaks down which external packages they use in how many different internal packages with stacked bars.

Note that every external dependency is counted separately so that if on internal package uses two external packages it will be displayed for both and so stacked twice.

```
<Figure size 640x480 with 0 Axes>
```


External package usage per artifact
number of packages
0
10
20
30
40
50
60
70
80
artifact
axon-messaging-5.0.2
axon-spring-boot-autoconfigure-5.0.2
axon-server-connector-5.0.2
axon-common-5.0.2
axon-conversion-5.0.2
axon-test-5.0.2
axon-eventsourcing-5.0.2
axon-modelling-5.0.2
axon-update-5.0.2
axon-tracing-opentelemetry-5.0.2
axon-metrics-micrometer-5.0.2
IdTypeParameterResolver
ScannedEntityCreator
WrappedEventCriteriaBuilderMethod
com.fasterxml.jackson.annotation
com.fasterxml.jackson.core
com.fasterxml.jackson.databind
com.fasterxml.jackson.databind.node
com.fasterxml.jackson.dataformat.cbor.databind
com.google.gson
com.google.protobuf
io.axoniq.axonserver.connector
io.axoniq.axonserver.connector.admin
io.axoniq.axonserver.connector.command
io.axoniq.axonserver.connector.control
io.axoniq.axonserver.connector.event
io.axoniq.axonserver.connector.impl
io.axoniq.axonserver.connector.query
io.axoniq.axonserver.grpc
io.axoniq.axonserver.grpc.command
io.axoniq.axonserver.grpc.control
io.axoniq.axonserver.grpc.event
io.axoniq.axonserver.grpc.event.dcb
io.axoniq.axonserver.grpc.query
io.grpc
io.grpc.netty.shaded.io.grpc.netty
io.grpc.netty.shaded.io.netty.handler.ssl
io.grpc.stub
io.micrometer.core.instrument
io.micrometer.core.instrument.simple
io.opentelemetry.api
io.opentelemetry.api.trace
io.opentelemetry.context
io.opentelemetry.context.propagation
jakarta.annotation
jakarta.persistence
jakarta.validation
javax.annotation
javax.cache
javax.cache.configuration
javax.cache.event
org.apache.avro
org.apache.avro.generic
org.apache.avro.io
org.apache.avro.message
org.apache.avro.specific
org.apache.commons.lang3.tuple
org.awaitility
org.awaitility.core
org.axonframework.extension.spring
org.axonframework.extension.spring.authorization
org.axonframework.extension.spring.config
org.axonframework.extension.spring.conversion.avro
org.axonframework.extension.spring.messaging.unitofwork
org.ehcache.config
org.ehcache.core
org.ehcache.event
org.hamcrest
org.jetbrains.annotations
org.junit.jupiter.api
org.reactivestreams
org.slf4j
org.springframework.beans
org.springframework.beans.factory
org.springframework.beans.factory.config
org.springframework.beans.factory.support
org.springframework.boot.actuate.health
org.springframework.boot.autoconfigure
org.springframework.boot.autoconfigure.condition
org.springframework.boot.autoconfigure.orm.jpa
org.springframework.boot.autoconfigure.service.connection

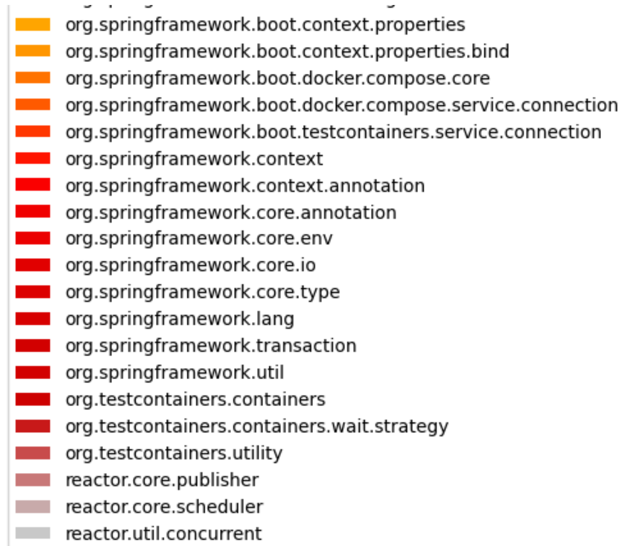


Table 7 Chart 2 - Top 15 external dependency using artifacts and their external packages (first 2 levels) stacked

The following chart shows the top 15 external package using artifacts and breaks down which external packages (first 2 levels) are used in how many different internal packages with stacked bars.

Note that every external dependency is counted separately so that if on internal package uses two external packages it will be displayed for both and so stacked twice.

```
<Figure size 640x480 with 0 Axes>
```

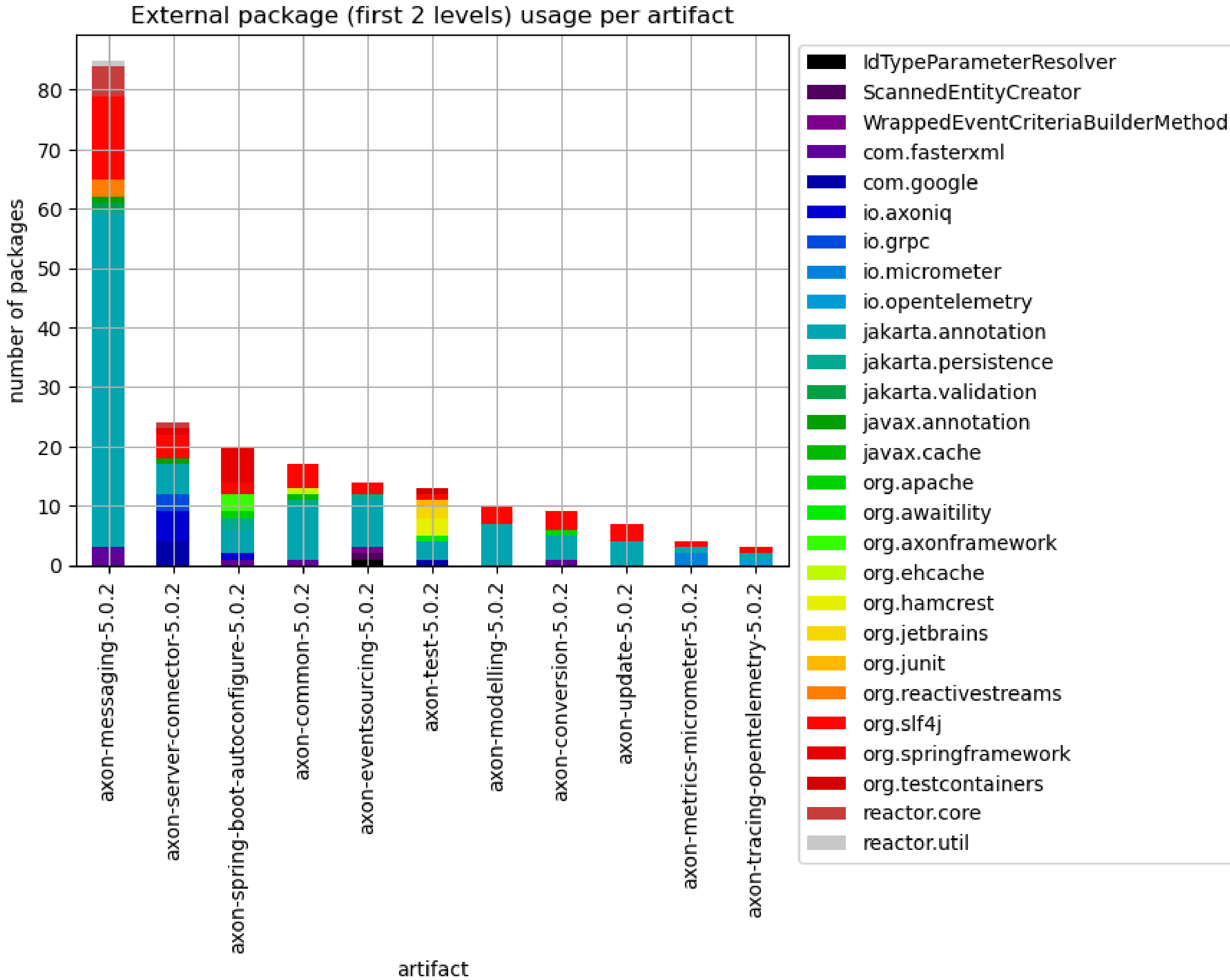


Table 8 - External usage per artifact

The following table shows the most used external packages separately for each artifact including external annotations. The results are grouped per artifact and sorted by the artifacts with the highest external type usage rate descending. Additionally, for each artifact the top 5 used external packages are listed in the top5ExternalPackages column.

The intention of this table is to find artifacts that use a lot of external dependencies in relation to their size and get an overview per artifact with the top 5 used external packages, the number of external types and packages used etc. .

Only the last 40 entries are shown. The whole table can be found in the following CSV report: `External_package_usage_per_artifact_sorted_top`

Columns:

- *artifactName* is used to group the the external package usage per artifact for a more detailed analysis.
- *numberOfTypesInArtifact* represents the total count of all analyzed types for the artifact

- *numberOfExternalTypesInArtifact* is the number of all external types that are used by the artifact
- *numberOfExternalPackagesInArtifact* is the number of all external packages that are used by the artifact
- *externalTypeRate* is the numberOfExternalTypesInArtifact / numberOfTypesInArtifact * 100
- *numberOfExternalTypeCaller* refers to the distinct types that make use of the external package
- *numberOfExternalTypeCalls* includes every invocation or reference to the types in the external package
- *numberOfExternalPackages* is the number of distinct external packages used by the artifact
- *top5ExternalPackages* contains a list of the top 5 most used external packages of the artifact
- *someExternalTypes* contains a list of lists and is also mean't to provide some examples of external types used

	artifactName	numberOfTypesInArtifact	numberOfExternalTypesInArtifact	numberOfExternalPackagesInArtifact	externalTypeRate	number(
0	axon-tracing-opentelemetry-5.0.2	5	16	6	320.000000	
1	axon-server-connector-5.0.2	72	113	23	156.944444	
2	axon-spring-boot-autoconfigure-5.0.2	72	95	37	131.944444	
3	axon-conversion-5.0.2	30	32	11	106.666667	
4	axon-metrics-micrometer-5.0.2	13	13	4	100.000000	
5	axon-test-5.0.2	73	24	11	32.876712	
6	axon-eventsourcing-5.0.2	104	22	6	21.153846	
7	axon-common-5.0.2	156	29	11	18.589744	
8	axon-update-5.0.2	23	4	2	17.391304	
9	axon-messaging-5.0.2	570	33	9	5.789474	
10	axon-modelling-5.0.2	93	4	2	4.301075	

Table 9 - External usage per artifact and package

This table lists internal packages and the artifacts they belong to that use many different external types of a specific external package without taking external annotations into account.

Only the last 40 entries are shown. The whole table can be found in the following CSV report: `External_package_usage_per_artifact_and_package`

Columns:

- *artifactName* that contains the type that calls the external package
- *fullPackageName* is the package within the artifact that contains the type that calls the external package
- *externalPackageName* identifies the external package as described above
- *numberOfExternalTypeCaller* refers to the distinct types that make use of the external package
- *numberOfExternalTypeCalls* includes every invocation or reference to the types in the external package
- *numberOfTypesInPackage* represents the total count of all types in that package
- *externalTypeNames* contains a list of actually utilized types of the external package
- *packageName* contains the name of the package (last part of *fullPackageName*)

	artifactName	fullPackageName	externalPackageName	numberOfExternalTypeCaller	numberO
0	axon-metrics-micrometer-5.0.2	org.axonframework.extension.metrics.micrometer	io.micrometer.core.instrument	37	
1	axon-test-5.0.2	org.axonframework.test.matchers	org.hamcrest	36	
2	axon-server-connector-5.0.2	org.axonframework.axonserver.connector.event	io.axoniq.axonserver.grpc.event.dcb	28	
3	axon-server-connector-5.0.2	org.axonframework.axonserver.connector.util	io.grpc	17	
4	axon-conversion-5.0.2	org.axonframework.conversion.avro	org.apache.avro	16	
5	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.proc...	org.slf4j	15	
6	axon-server-connector-5.0.2	org.axonframework.axonserver.connector.query	io.axoniq.axonserver.grpc.query	14	
7	axon-eventsourcing-5.0.2	org.axonframework.eventsourcing.eventstore.jpa	jakarta.persistence	12	
8	axon-server-connector-5.0.2	org.axonframework.axonserver.connector	io.axoniq.axonserver.grpc	12	
9	axon-server-connector-5.0.2	org.axonframework.axonserver.connector.event	org.slf4j	10	
10	axon-server-connector-5.0.2	org.axonframework.axonserver.connector.query	io.axoniq.axonserver.connector	10	
11	axon-server-connector-5.0.2	org.axonframework.axonserver.connector.query	io.axoniq.axonserver.grpc	9	
12	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension.springboot.autoconfig	org.axonframework.extension.spring.config	9	
13	axon-tracing-opentelemetry-5.0.2	org.axonframework.extension.tracing.opentelemetry	io.opentelemetry.context.propagation	9	
14	axon-tracing-opentelemetry-5.0.2	org.axonframework.extension.tracing.opentelemetry	io.opentelemetry.api.trace	9	
15	axon-common-5.0.2	org.axonframework.common.caching	org.ehcache.event	8	
16	axon-common-5.0.2	org.axonframework.common.caching	javax.cache.event	8	
17	axon-conversion-5.0.2	org.axonframework.conversion.avro	org.apache.avro.message	8	
18	axon-conversion-5.0.2	org.axonframework.conversion.json	com.fasterxml.jackson.databind	8	
19	axon-messaging-5.0.2	org.axonframework.messaging.core.annotation	org.slf4j	8	
20	axon-server-connector-5.0.2	org.axonframework.axonserver.connector	io.axoniq.axonserver.grpc.control	8	
21	axon-server-connector-5.0.2	org.axonframework.axonserver.connector.command	io.axoniq.axonserver.grpc.command	8	
22	axon-server-connector-5.0.2	org.axonframework.axonserver.connector.event	io.axoniq.axonserver.connector.event	8	
23	axon-server-connector-5.0.2	org.axonframework.axonserver.connector.util	io.axoniq.axonserver.grpc	8	
24	axon-update-5.0.2	org.axonframework.update	org.slf4j	8	
25	axon-conversion-5.0.2	org.axonframework.conversion.avro	org.apache.avro.generic	7	
26	axon-eventsourcing-5.0.2	org.axonframework.eventsourcing.eventstore.jpa	org.slf4j	7	

	artifactName	fullPackageName	externalPackageName	numberOfExternalTypeCaller	numberO
27	axon-messaging-5.0.2	org.axonframework.messaging.core	reactor.core.publisher	7	
28	axon-server-connector-5.0.2	org.axonframework.axonserver.connector.command	io.axoniq.axonserver.grpc	7	
29	axon-test-5.0.2	org.axonframework.test.fixture	org.hamcrest	7	
30	axon-server-connector-5.0.2	org.axonframework.axonserver.connector.query	io.axoniq.axonserver.connector.query	6	
31	axon-common-5.0.2	org.axonframework.common	org.slf4j	5	
32	axon-common-5.0.2	org.axonframework.common.configuration	org.slf4j	5	
33	axon-common-5.0.2	org.axonframework.common.jpa	jakarta.persistence	5	
34	axon-messaging-5.0.2	org.axonframework.messaging.core	org.reactivestreams	5	
35	axon-messaging-5.0.2	org.axonframework.messaging.core.interception	jakarta.validation	5	
36	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.proc...	jakarta.persistence	5	
37	axon-server-connector-5.0.2	org.axonframework.axonserver.connector	io.axoniq.axonserver.connector	5	
38	axon-server-connector-5.0.2	org.axonframework.axonserver.connector.event	io.axoniq.axonserver.grpc.control	5	
39	axon-server-connector-5.0.2	org.axonframework.axonserver.connector.event	io.axoniq.axonserver.connector	5	

Table 10 - Top 20 external package usage per type

This table shows internal types that utilize the most different external types and packages. These have the highest probability of change depending on external libraries. A case-by-case approach is also advisable here because there could for example also be code units that encapsulate an external library and have this high count of external dependencies on purpose.

Only the last 20 entries are shown. The whole table can be found in the following CSV report: `External_package_usage_per_type`

Columns:

- *artifactName* that contains the type that calls the external package
- *fullPackageName* is the package within the artifact that contains the type that calls external types
- *typeName* identifies the internal type within the package and artifact that calls external types
- *numberOfExternalTypeCaller* and *numberOfExternalTypes* refers to the distinct external types that are used by the internal type
- *numberOfExternalTypeCalls* includes every invocation or reference to the types in the external package
- *numberOfTypesInPackage* represents the total count of all types in that package

- *numberOfExternalPackages* shows how many different external packages are used by the internal type
- *externalPackageNames* contains the list of names of the different external packages that are used by the internal type
- *externalTypeNames* contains a list of actually utilized types of the external package
- *packageName* contains the name of the package (last part of *fullPackageName*)

	artifactName	fullPackageName	typeName	numberOfExternalTypeCaller	numb
0	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension.springboot.autoconfig	ConverterAutoConfiguration	16	
1	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension.springboot.autoconfig	AxonServerAutoConfiguration	14	
2	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension.springboot.autoconfig	AvroSchemaStoreAutoConfiguration	14	
3	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension.springboot.util	AbstractQualifiedBeanCondition	12	
4	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension.springboot.autoconfig	EventProcessingAutoConfiguration	10	
5	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension.springboot.autoconfig	JpaEventStoreAutoConfiguration	8	
6	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension.springboot.autoconfig	JpaTransactionAutoConfiguration	8	
7	axon-server-connector-5.0.2	org.axonframework.axonserver.connector	AxonServerConnectionManager$Builder	9	
8	axon-server-connector-5.0.2	org.axonframework.axonserver.connector.event	AggregateBasedAxonServerEventStorageEngine	13	
9	axon-server-connector-5.0.2	org.axonframework.axonserver.connector.event	EventProcessorControlService	7	
10	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension.springboot.autoconfig	JpaAutoConfiguration	8	
11	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension.springboot.autoconfig	JdbcTransactionAutoConfiguration	7	
12	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension.springboot.util	DefaultEntityRegistrar	6	
13	axon-conversion-5.0.2	org.axonframework.conversion.avro	SpecificRecordBaseConverterStrategy	8	
14	axon-conversion-5.0.2	org.axonframework.conversion.avro	AvroUtil	16	
15	axon-server-connector-5.0.2	org.axonframework.axonserver.connector.command	AxonServerCommandBusConnector	9	
16	axon-server-connector-5.0.2	org.axonframework.axonserver.connector.event	AxonServerEventStorageEngine	12	
17	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension.springboot.autoconfig	AxonTimeoutAutoConfiguration	5	
18	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension.springboot.autoconfig	CBORMapperAutoConfiguration	8	
19	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension.springboot.autoconfig	ObjectMapperAutoConfiguration	9	

Table 11 - External package usage distribution per type

This table shows how many types use one external package, how many use two, etc. . This gives an overview of the distribution of external package calls and the overall coupling to external libraries. The higher the count of distinct external packages the lower should be the count of types that use them. Dependencies to external annotations are left out here.

More details about which types have the highest external package dependency usage can be in the tables 4 and 5 above.

Only the last 40 entries are shown. The whole table can be found in the following CSV report: `External_package_usage_per_artifact_distribution`

Columns:

- *artifactName* that contains the type that calls the external package
- *artifactTypes* the total count of types in the artifact
- *numberOfExternalPackages* the number of distinct external packages used
- *numberOfTypes* in the artifact where the *numberOfExternalPackages* applies
- *numberOfTypesPercentage* in the artifact where the *numberOfExternalPackages* applies in %

	artifactName	artifactPackages	artifactTypes	numberOfExternalPackages	numberOfPackages	numberOfTypes	typesCallingExternalRate
0	axon-messaging-5.0.2	57	570	7	18	45	7.894737
1	axon-common-5.0.2	15	156	10	7	18	11.538462
2	axon-spring-boot-autoconfigure-5.0.2	7	72	33	7	34	47.222222
3	axon-server-connector-5.0.2	5	72	20	5	51	70.833333
4	axon-eventsourcing-5.0.2	8	104	5	4	11	10.576923
5	axon-test-5.0.2	5	73	9	4	24	32.876712
6	axon-conversion-5.0.2	4	30	10	3	15	50.000000
7	axon-modelling-5.0.2	7	93	1	3	3	3.225806
8	axon-update-5.0.2	5	23	1	3	7	30.434783
9	axon-metrics-micrometer-5.0.2	2	13	3	2	12	92.307692
10	axon-tracing-opentelemetry-5.0.2	1	5	5	1	5	100.000000

Table 12 - External package usage per artifact grouped by number of internal packages

The following table shows the external package usage for every artifact grouped by the number of distinct internal dependent packages. The intention is to find external package usage spread across multiple internal packages in artifacts.

Artifacts that encapsulate external dependency calls in one internal package overall (or each) are easier to change if those external dependencies change and are most likely applying a Hexagonal architecture. Artifacts that use external dependencies in multiple internal packages need more effort to adapt to changes of those external dependencies. On one hand this could be intended e.g. when using standardized libraries. On the other hand this might indicate higher than necessary coupling.

The whole table can be found in the following CSV report: `External_package_usage_per_internal_package_count`

artifactName	axon-common-5.0.2	axon-conversion-5.0.2	axon-eventsourcing-5.0.2	axon-messaging-5.0.2	axon-metrics-micrometer-5.0.2	axon-modelling-5.0.2	axon-server-connector-5.0.2	axon-spring-boot-autoconfigure-5.0.2	axon-test-5.0.2	axon-update-5.0.2
numberOfPackages										
2	0.000000	0.0	25.0	0.000000	100.0	0.000000	40.0	28.571429	40.0	0.0
3	0.000000	75.0	0.0	5.263158	0.0	42.857143	60.0	42.857143	60.0	60.0
4	26.666667	100.0	0.0	0.000000	0.0	0.000000	80.0	57.142857	0.0	80.0
5	0.000000	0.0	0.0	8.771930	0.0	0.000000	100.0	0.000000	0.0	0.0
7	0.000000	0.0	0.0	0.000000	0.0	100.000000	0.0	0.000000	0.0	0.0
8	0.000000	0.0	100.0	0.000000	0.0	0.000000	0.0	0.000000	0.0	0.0
9	60.000000	0.0	0.0	0.000000	0.0	0.000000	0.0	0.000000	0.0	0.0
14	0.000000	0.0	0.0	24.561404	0.0	0.000000	0.0	0.000000	0.0	0.0
56	0.000000	0.0	0.0	98.245614	0.0	0.000000	0.0	0.000000	0.0	0.0

Table 13 - External package usage aggregated

This table lists all artifacts and their external package dependencies usage aggregated over internal packages.

The intention behind this is to find artifacts that use an external dependency across multiple internal packages. This might be intended for frameworks and standardized libraries and helps to quantify how widely those are used. For some external dependencies it might be beneficial to only access it from one package and provide an abstraction for internal usage following a Hexagonal architecture. Thus, this table may also help in finding application for the Hexagonal architecture or similar approaches (Domain Driven Design Anti Corruption Layer). After all it is easier to update or replace such external dependencies when they are used in specific areas and not all over the code.

Only the last 40 entries are shown. The whole table can be found in the following CSV report: `External_package_usage_per_artifact_package_aggregated`

Columns:

- *artifactName* that contains the type that calls the external package
- *artifactPackages* is the total count of packages in the artifact
- *artifactTypes* is the total count of types in the artifact
- *numberOfExternalPackages* the number of distinct external packages used

- *[min,max,med,avg,std]NumberOfPackages* provide statistics based on each external package and its package usage within the artifact
- *[min,max,med,avg,std]NumberOfPackagesPercentage* provide statistics in % based on each external package and its package usage within the artifact in respect to the overall count of packages in the artifact
- *[min,max,med,avg,std]NumberOfTypes* provide statistics based on each external package and its type usage within the artifact
- *[min,max,med,avg,std]NumberOfTypePercentage* provide statistics in % based on each external package and its type usage within the artifact in respect to the overall count of packages in the artifact
- *numberOfTypes* in the artifact where the *numberOfExternalPackages* applies
- *numberOfTypesPercentage* in the artifact where the *numberOfExternalPackages* applies in %

Table 13a - External package usage aggregated - count of internal packages

	artifactName	artifactPackages	numberOfExternalPackages	minNumberOfPackages	medNumberOfPackages	avgNumberOfPackages	maxN
0	axon-messaging-5.0.2	57	7	1	1.0	3.714286	
1	axon-server-connector-5.0.2	5	20	1	1.0	1.900000	
2	axon-common-5.0.2	15	10	1	1.0	1.300000	
3	axon-conversion-5.0.2	4	10	1	1.0	1.200000	
4	axon-modelling-5.0.2	7	1	3	3.0	3.000000	
5	axon-spring-boot-autoconfigure-5.0.2	7	33	1	1.0	1.303030	
6	axon-test-5.0.2	5	9	1	1.0	1.222222	
7	axon-update-5.0.2	5	1	3	3.0	3.000000	
8	axon-eventsourcing-5.0.2	8	5	1	1.0	1.200000	
9	axon-metrics-micrometer-5.0.2	2	3	1	1.0	1.333333	
10	axon-tracing-opentelemetry-5.0.2	1	5	1	1.0	1.000000	

Table 13b - External package usage aggregated - percentage of internal packages

	artifactName	artifactPackages	numberOfExternalPackages	minNumberOfPackagesPercentage	medNumberOfPackagesPercentage	avgNur
0	axon-messaging-5.0.2	57	7	1.754386	1.754386	
1	axon-server-connector-5.0.2	5	20	20.000000	20.000000	
2	axon-common-5.0.2	15	10	6.666667	6.666667	
3	axon-conversion-5.0.2	4	10	25.000000	25.000000	
4	axon-modelling-5.0.2	7	1	42.857143	42.857143	
5	axon-spring-boot-autoconfigure-5.0.2	7	33	14.285714	14.285714	
6	axon-test-5.0.2	5	9	20.000000	20.000000	
7	axon-update-5.0.2	5	1	60.000000	60.000000	
8	axon-eventsourcing-5.0.2	8	5	12.500000	12.500000	
9	axon-metrics-micrometer-5.0.2	2	3	50.000000	50.000000	
10	axon-tracing-opentelemetry-5.0.2	1	5	100.000000	100.000000	

Table 13c - External package usage aggregated - count of internal types

	artifactName	artifactTypes	numberOfExternalPackages	minNumberOfTypes	medNumberOfTypes	avgNumberOfTypes	maxNumberOfTypes
0	axon-messaging-5.0.2	570	7	2	2.0	7.428571	30
1	axon-server-connector-5.0.2	72	20	1	3.5	5.200000	18
2	axon-common-5.0.2	156	10	1	1.0	2.300000	8
3	axon-conversion-5.0.2	30	10	1	2.5	3.100000	6
4	axon-modelling-5.0.2	93	1	3	3.0	3.000000	3
5	axon-spring-boot-autoconfigure-5.0.2	72	33	1	2.0	2.272727	7
6	axon-test-5.0.2	73	9	1	1.0	3.333333	22
7	axon-update-5.0.2	23	1	7	7.0	7.000000	7
8	axon-eventsourcing-5.0.2	104	5	1	1.0	3.000000	6
9	axon-metrics-micrometer-5.0.2	13	3	1	1.0	4.666667	12
10	axon-tracing-opentelemetry-5.0.2	5	5	1	2.0	2.200000	4

Table 13d - External package usage aggregated - percentage of internal types

	artifactName	artifactTypes	numberOfExternalPackages	minNumberOfTypesPercentage	medNumberOfTypesPercentage	avgNumberOfType
0	axon-messaging-5.0.2	570	7	0.350877	0.350877	
1	axon-server-connector-5.0.2	72	20	1.388889	4.861111	
2	axon-common-5.0.2	156	10	0.641026	0.641026	
3	axon-conversion-5.0.2	30	10	3.333333	8.333333	
4	axon-modelling-5.0.2	93	1	3.225806	3.225806	
5	axon-spring-boot-autoconfigure-5.0.2	72	33	1.388889	2.777778	
6	axon-test-5.0.2	73	9	1.369863	1.369863	
7	axon-update-5.0.2	23	1	30.434783	30.434783	
8	axon-eventsourcing-5.0.2	104	5	0.961538	0.961538	
9	axon-metrics-micrometer-5.0.2	13	3	7.692308	7.692308	
10	axon-tracing-opentelemetry-5.0.2	5	5	20.000000	40.000000	

Table 13 Chart 1 - External package usage - max percentage of internal types

This chart shows per artifact the maximum percentage of internal packages (compared to all packages in that artifact) that use one specific external package.

Example: One artifact might use 10 external packages where 7 of them are used in one internal package, 2 of them are used in two packages and one external dependency is used in 5 packages. So for this artifact there will be a point at x = 10 (external packages used by the artifact) and 5 (max internal packages). Instead of the count the percentage of internal packages compared to all packages in that artifact is used to get a normalized plot.

```
<Figure size 640x480 with 0 Axes>
```

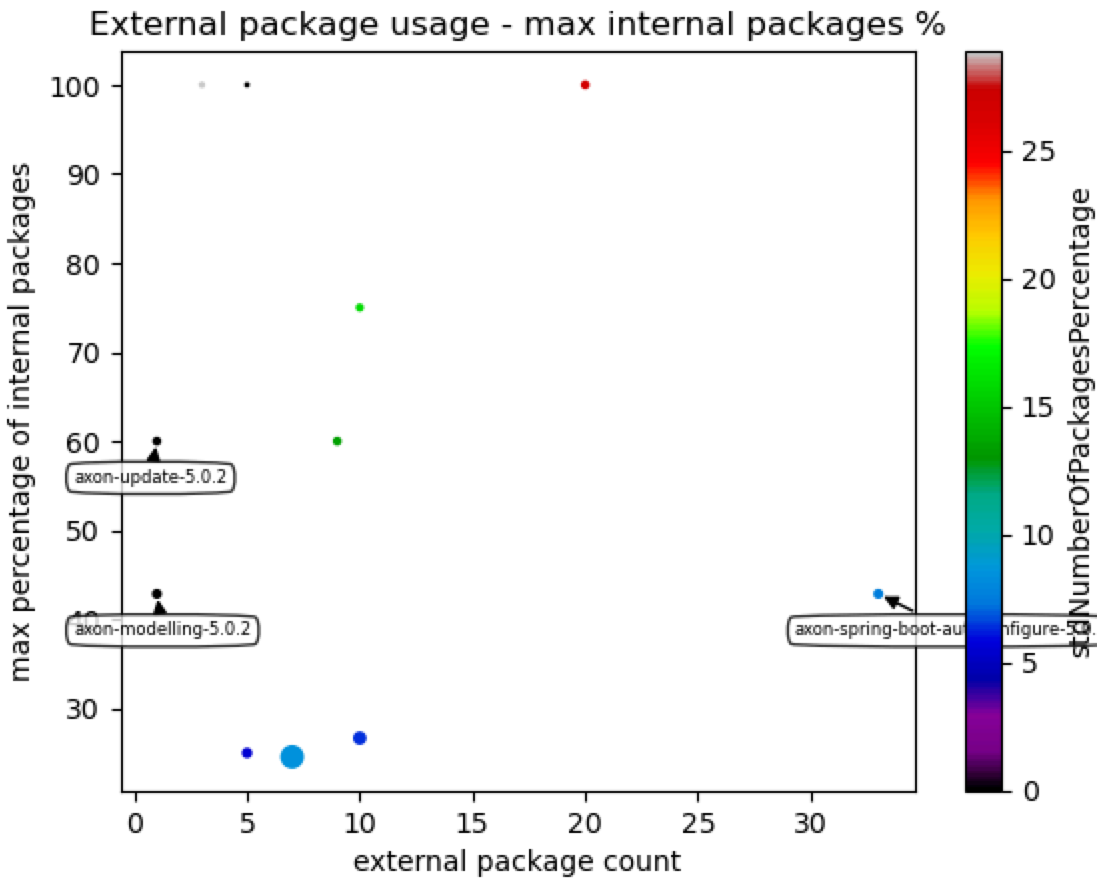
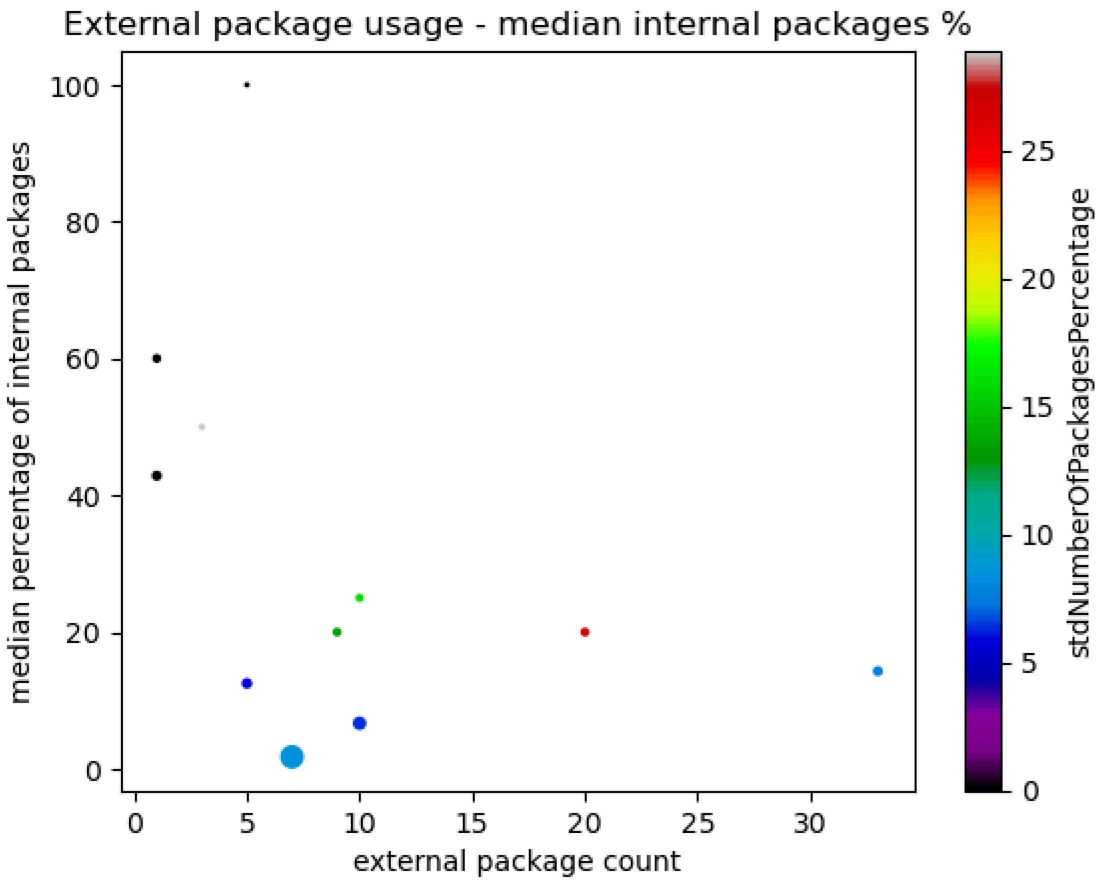


Table 13 Chart 2 - External package usage - median percentage of internal types

This chart shows per artifact the median (0.5 percentile) of internal packages (compared to all packages in that artifact) that use one specific external package.

Example: One artifact might use 9 external packages where 3 of them are used in 1 internal package, 3 of them are used in 2 package and the last 3 ones are used in 3 packages. So for this artifact there will be a point at x = 10 (external packages used by the artifact) and 2 (median internal packages). Instead of the count the percentage of internal packages compared to all packages in that artifact is used to get a normalized plot.

```
<Figure size 640x480 with 0 Axes>
```



Maven POMs

Table 14 - Maven POMs and their declared dependencies

If Maven is used as for package and dependency management and a ".pom" file is included in the artifact, the following table shows the external dependencies that are declared there.

	pom.artifactId	pom.name	scope	dependency.optional	dependentArtifact.group	dependentArtifact.name
0	axon-common	Axon Framework - Common	default	True	org.ehcache	ehcache
1	axon-common	Axon Framework - Common	default	False	com.fasterxml.jackson.core	jackson-core
2	axon-common	Axon Framework - Common	default	True	javax.cache	cache-api
3	axon-common	Axon Framework - Common	default	True	jakarta.validation	jakarta.validation-api
4	axon-common	Axon Framework - Common	default	False	com.fasterxml.jackson.core	jackson-databind
...	...	...	...	...	...	...
190	axon-tracing-opentelemetry	Axon Extension - Tracing - OpenTelemetry	default	False	jakarta.annotation	jakarta.annotation-api
191	axon-tracing-opentelemetry	Axon Extension - Tracing - OpenTelemetry	default	False	io.opentelemetry	opentelemetry-api
192	axon-update	Axon Framework - Update	test	False	org.axonframework	axon-common
193	axon-update	Axon Framework - Update	default	False	org.axonframework	axon-common
194	axon-update	Axon Framework - Update	default	False	jakarta.annotation	jakarta.annotation-api

195 rows × 6 columns